

CHEM 210

Modern Applications of Chemistry

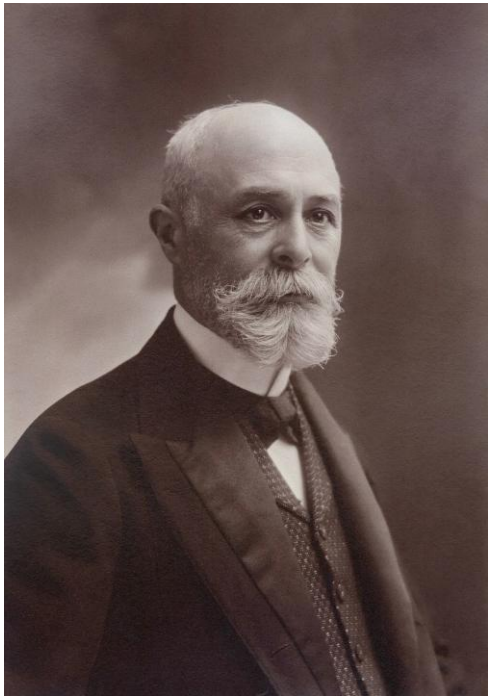
Duke University

Prof. Charlie Cox

Topics:

- 1.) An overview of the 20th century discoveries that were paramount in understanding nuclear and atomic structure.
- 2.) Nuclear Structure and the Strong Nuclear Force
- 3.) Einstein's Discovery and Mass Defect





Becquerel



1903

1896



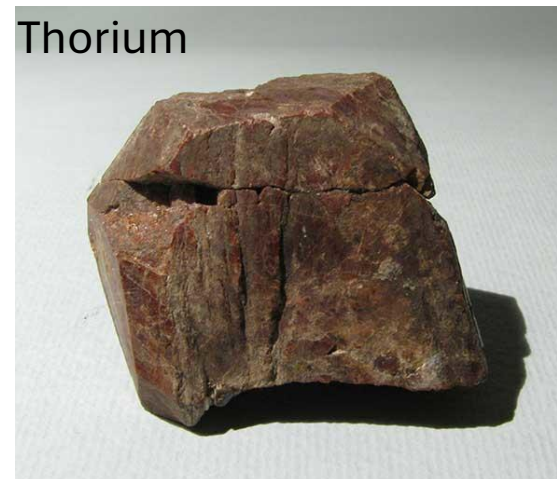
Uranium

Conclusion: Something is being emitted, but what?

Curie reached the conviction that the emission of rays by the compounds of uranium is a property of the metal itself—that it is an atomic property of the element uranium independent of its chemical or physical state



Curie



Thorium



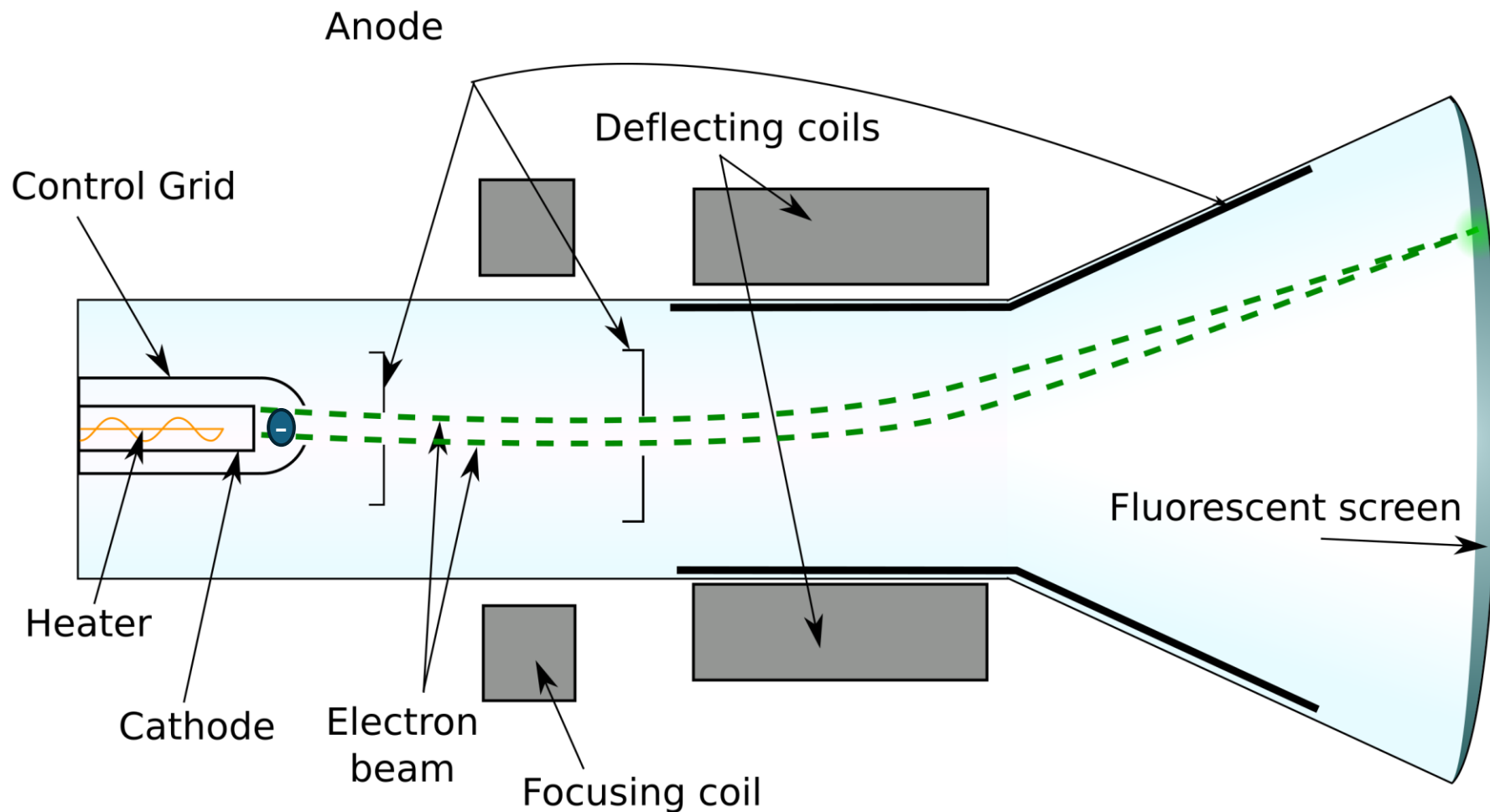
Radium

Cathode Rays, aka Electrons

$$F = \frac{q_- q_+}{kr^2}$$



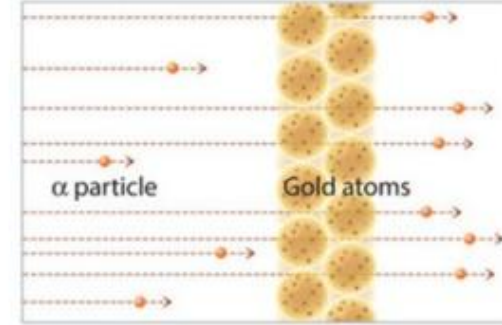
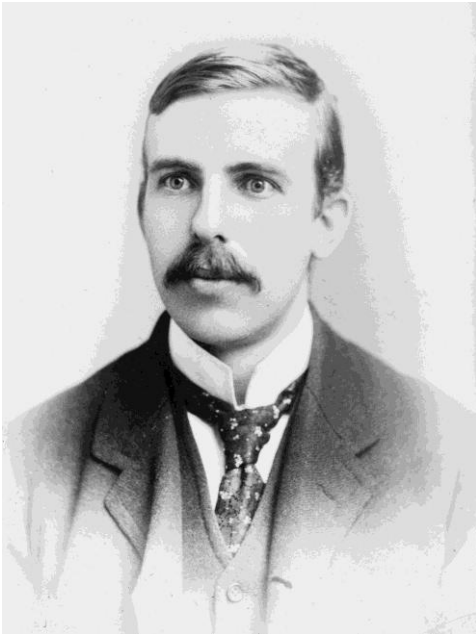
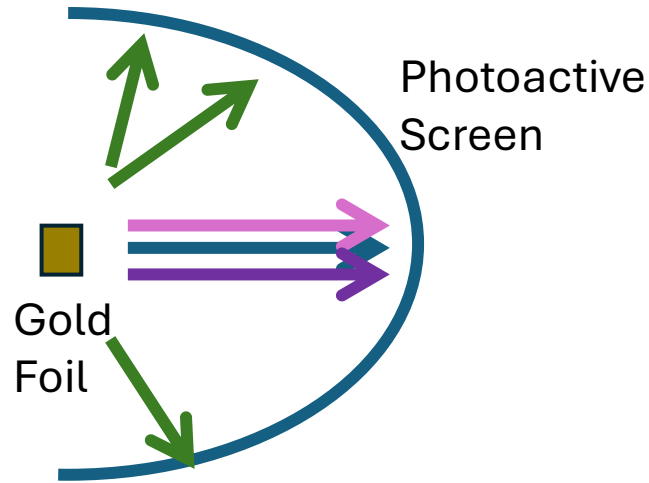
1897



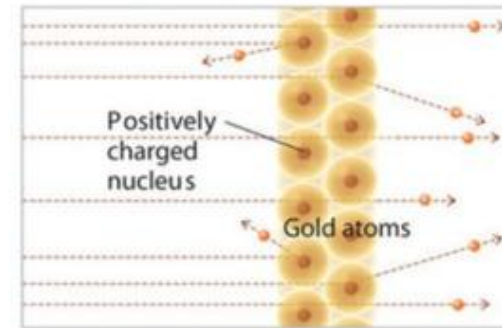
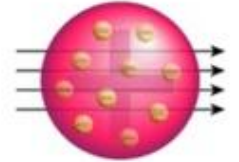
Lord Rutherford

Alpha
Emitter

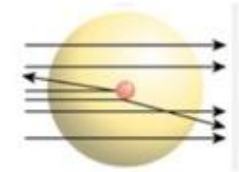
Positive
Charge



(b) What Rutherford expected if Thomson's model were correct



(c) What Rutherford actually observed



Gravimetric Analysis

22.98976928	24.305	IIIB	IVB	VB	VIB	VII B	VIII B
19 K Potassium 39.0983	20 Ca Calcium 40.078	21 Sc Scandium 44.955908	22 Ti Titanium 47.867	23 V Vanadium 50.9415	24 Cr Chromium 51.9961	25 Mn Manganese 54.938044	26 Fe Iron 55.845
37 Rb Rubidium 85.4678	38 Sr Strontium 87.62	39 Y Yttrium 88.90584	40 Zr Zirconium 91.224	41 Nb Niobium 92.90637	42 Mo Molybdenum 95.95	43 Tc Technetium (98)	44 Ru Ruthenium 101.07
55 Cs Caesium	56 Ba Barium	57 - 71 Lanthanoids	72 Hf Hafnium	73 Ta Tantalum	74 W Tungsten	75 Re Rhenium	76 Os Osmium

T.W. Richards sought to “weigh” the elements through a series of reactions. However, something weird happened....

Isotopes of Magnesium

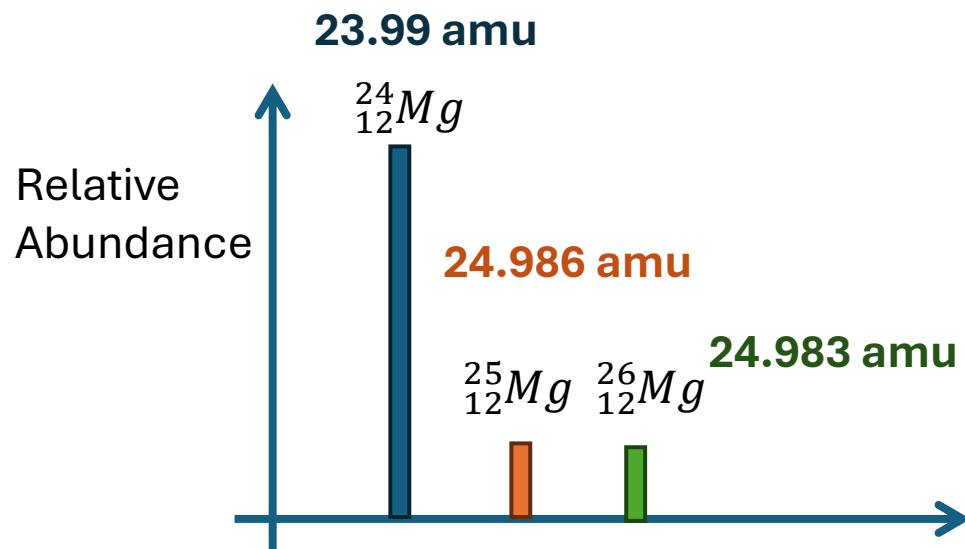


→
Dissolve MgCl_2 in
Water and Analyze
with Mass Spectrometry

Time of Flight Mass/Charge Separator



Detector

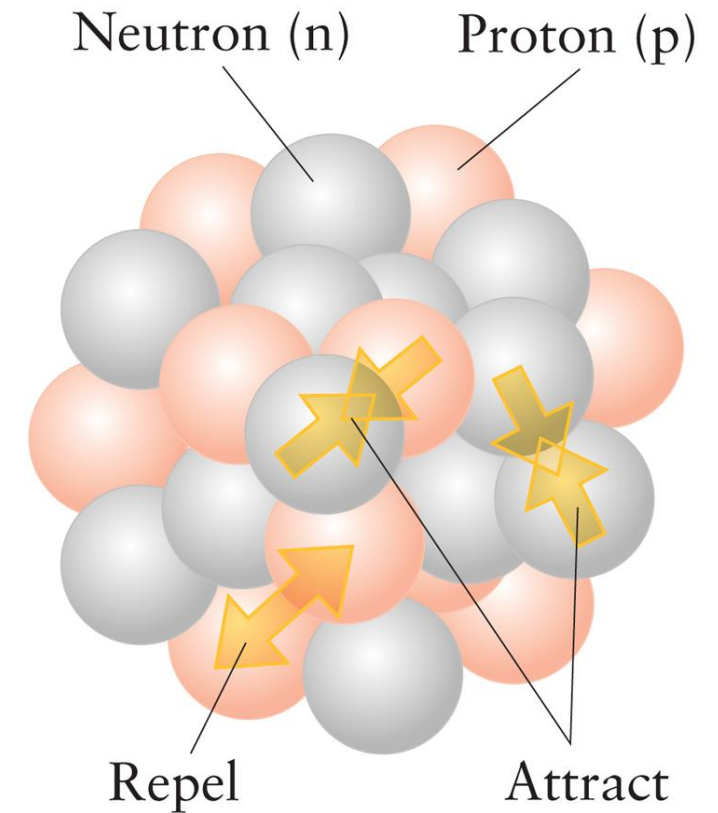


What holds together the Nucleus?

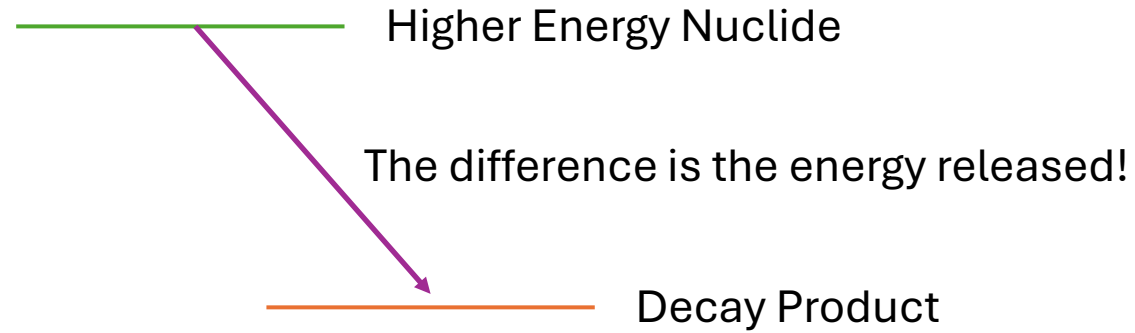
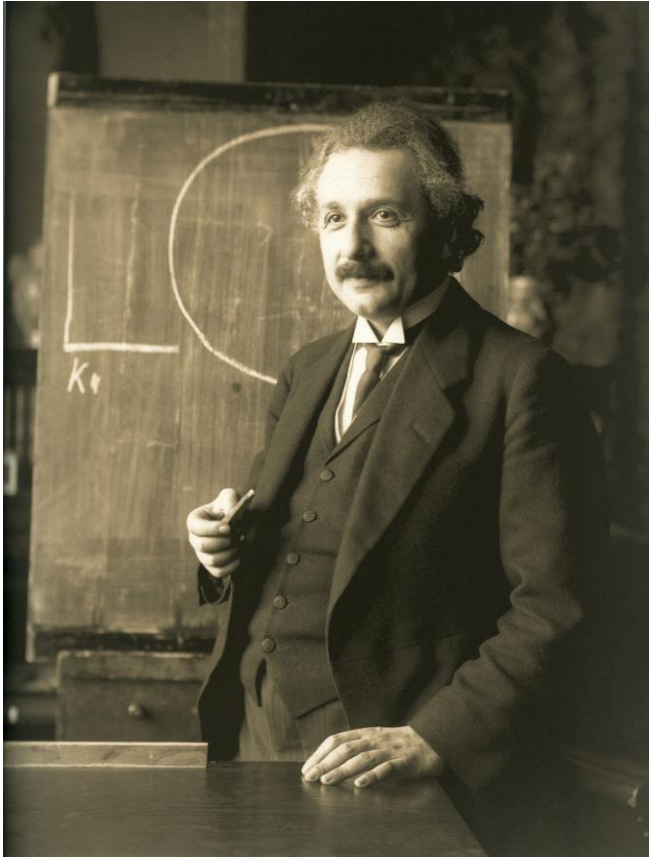
The particles in the nucleus are held together by a very strong attractive force found only in the nucleus called the **strong force**. Acts over only very short distances

Protons and neutrons are called **nucleons**.

The neutrons play an important role in stabilizing the nucleus, since they add to the strong force but don't repel each other like the protons do.



Einstein, Energy, and Mass



Calculate the mass defect in Fe-56 if the mass of an Fe-56 nucleus is 55.921 amu. The mass of a proton is 1.00728 amu, the mass of a neutron is 1.008665 amu, and the mass of an electron is 0.00054848 amu. Calculate the binding energy for Fe-56. $1 \text{ amu} = 1.66 \times 10^{-27} \text{ kg}$

Learning Outcomes

Explain the significance of mass defect and use the mass defect to calculate binding energy.

Extra Practice

How many years could a 100-Watt lightbulb be powered by the energy from the mass defect of ONE moles of Carbon-14? 1 Watt = 1 J/s

$$1.66 \times 10^{-27} \text{ kg} = 1 \text{ amu}$$

$$\text{Carbon} - 14 = 14.003241 \text{ amu}$$

$$\text{Mass of a Proton} = 1.6726231 \times 10^{-27} \text{ kg}$$

$$\text{Mass of a Neutron} = 1.6749 \times 10^{-27} \text{ kg}$$

$$\text{Mass of an Electron} = 9.1093837 \times 10^{-27} \text{ kg}$$

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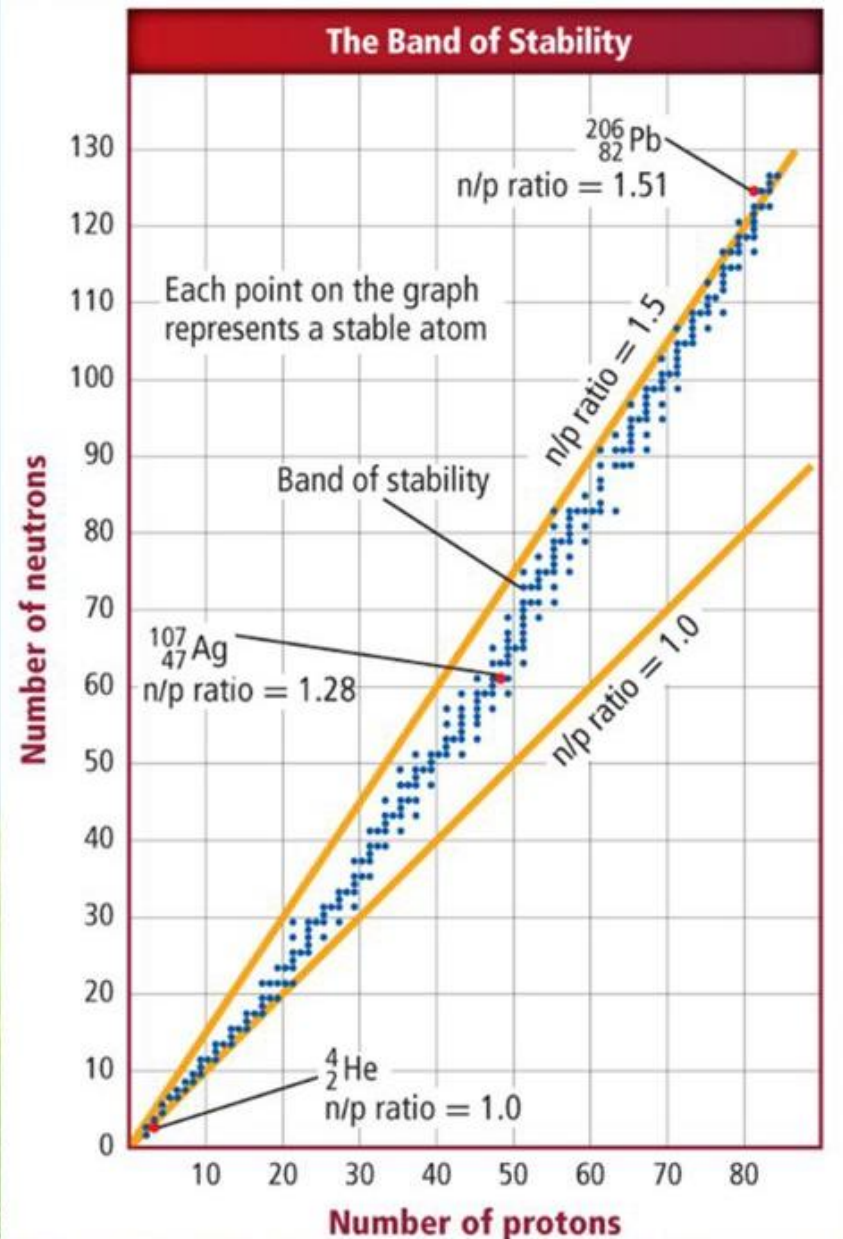
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Topics:

- 1.) Band of Stability
- 2.) Modes of Nuclear Decay
- 3.) Fission and Fusion Processes
- 4.) Nuclear Energy
- 5.) Kinetics of Nuclear Decay



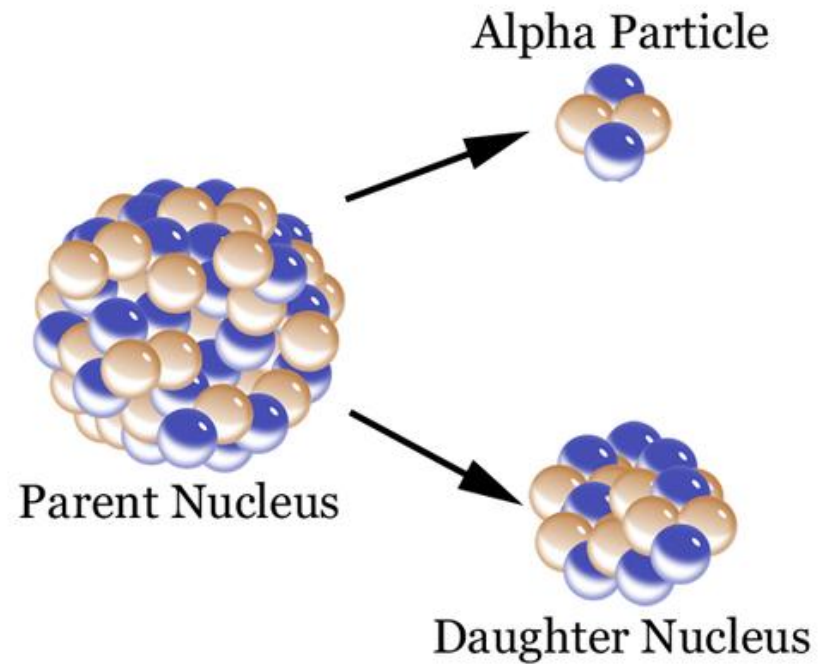
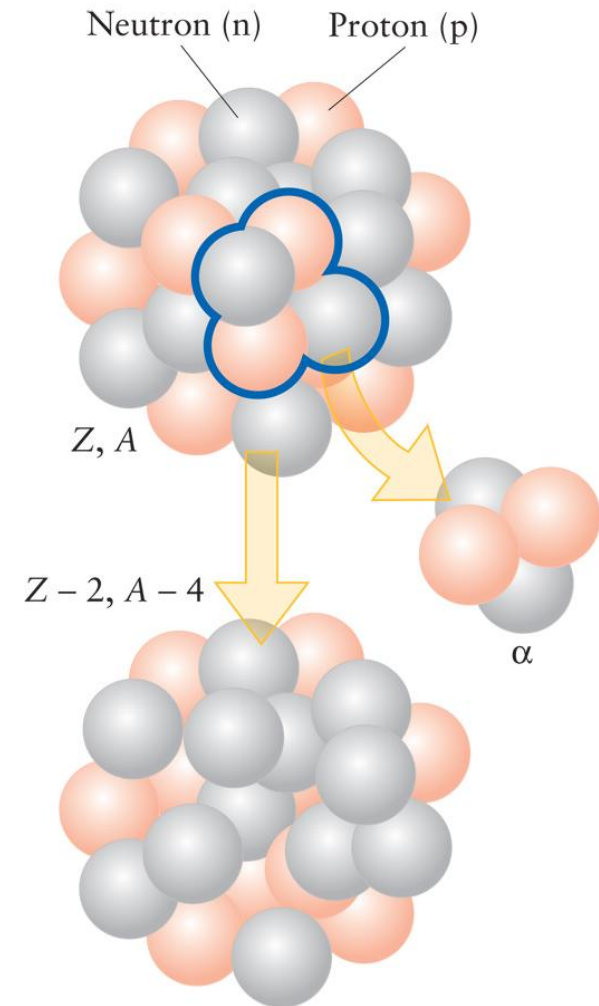
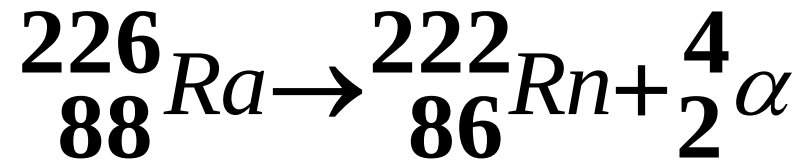
Band of Stability



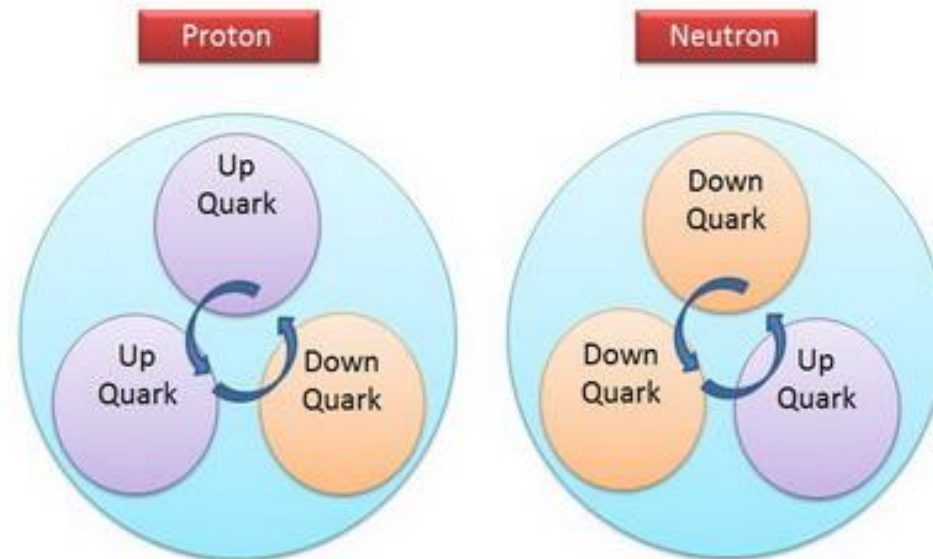
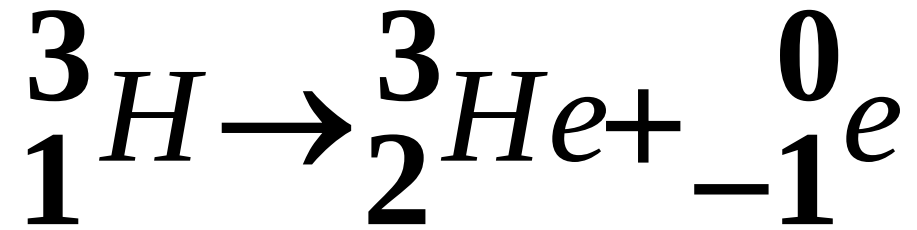
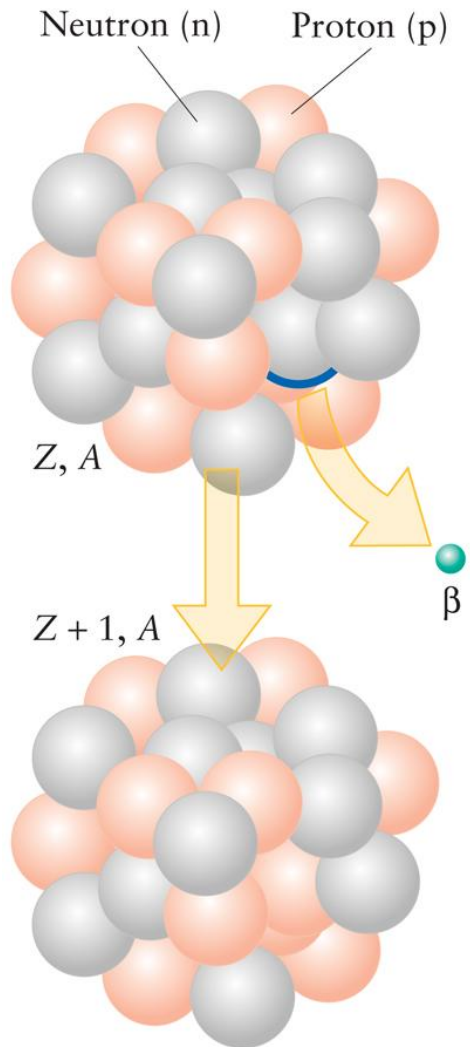
Number of Stable Isotopes	Nuclei Composition
166	Even Number of Protons and Even Number of Neutrons
55	
52	
6	Odd Number of Protons and Odd Number of Neutrons

80 Hg Mercury (200.592)	81 Tl Thallium (204.38)	82 Pb Lead (207.2)	83 Bi Bismuth (208.98040)	84 Po Polonium (209)	85 At Astatine (210)	86 Rn Radon (222)
112 Cn Copernicium (285)	113 Nh Nihonium (286)	114 Fl Flerovium (289)	115 Mc Moscovium (289)	116 Lv Livermorium (293)	117 Ts Tennessine (294)	118 Og Oganesson (294)
66 Dy Dysprosium (162.500)	67 Ho Holmium (164.93033)	68 Er Erbium (167.259)	69 Tm Thulium (168.93422)	70 Yb Ytterbium (173.045)	71 Lu Lutetium (174.9668)	
98 Cf Californium (251)	99 Es Einsteinium (252)	100 Fm Fermium (257)	101 Md Mendelevium (258)	102 No Nobelium (259)	103 Lr Lawrencium (260)	

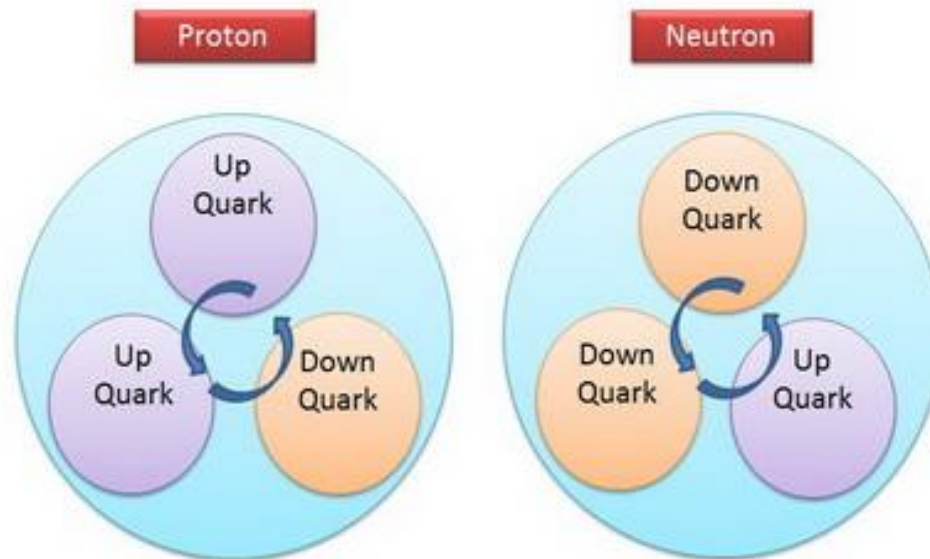
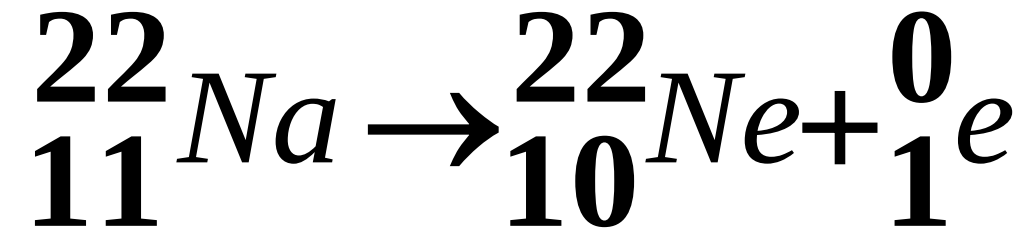
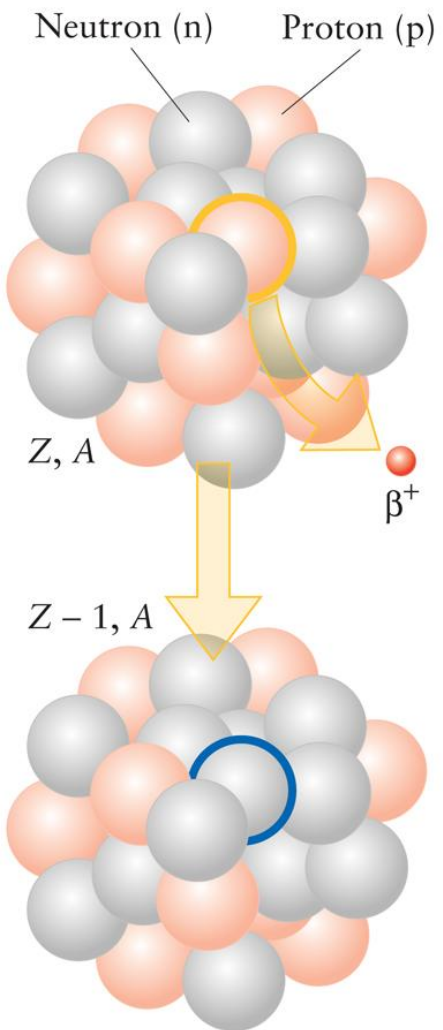
Alpha Decay



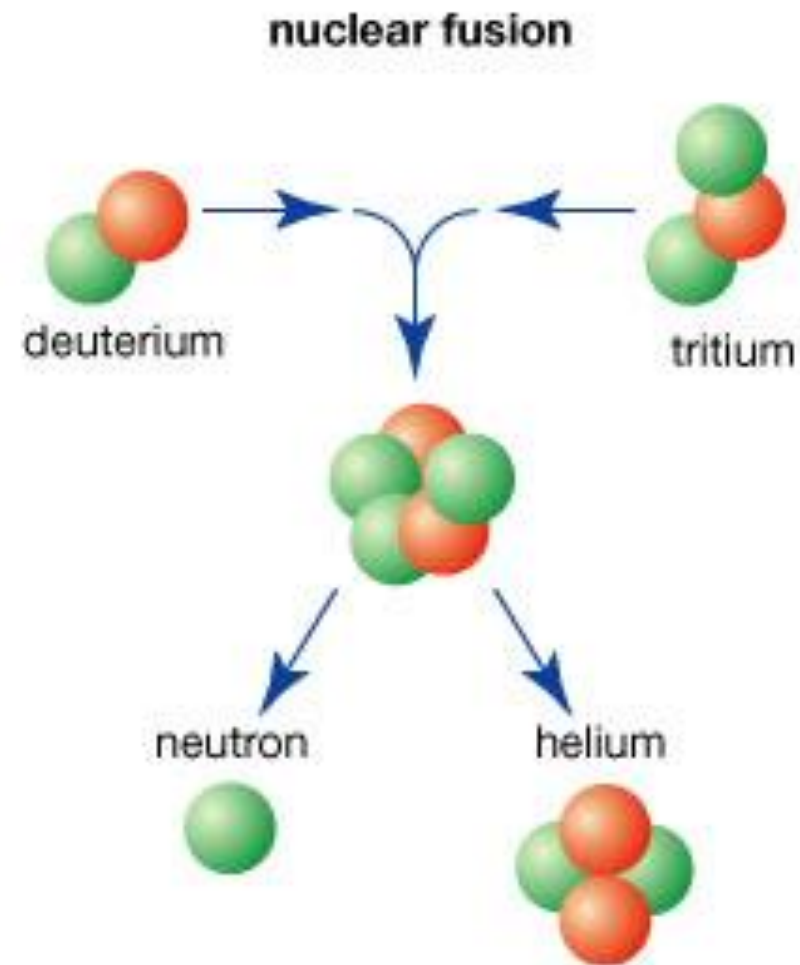
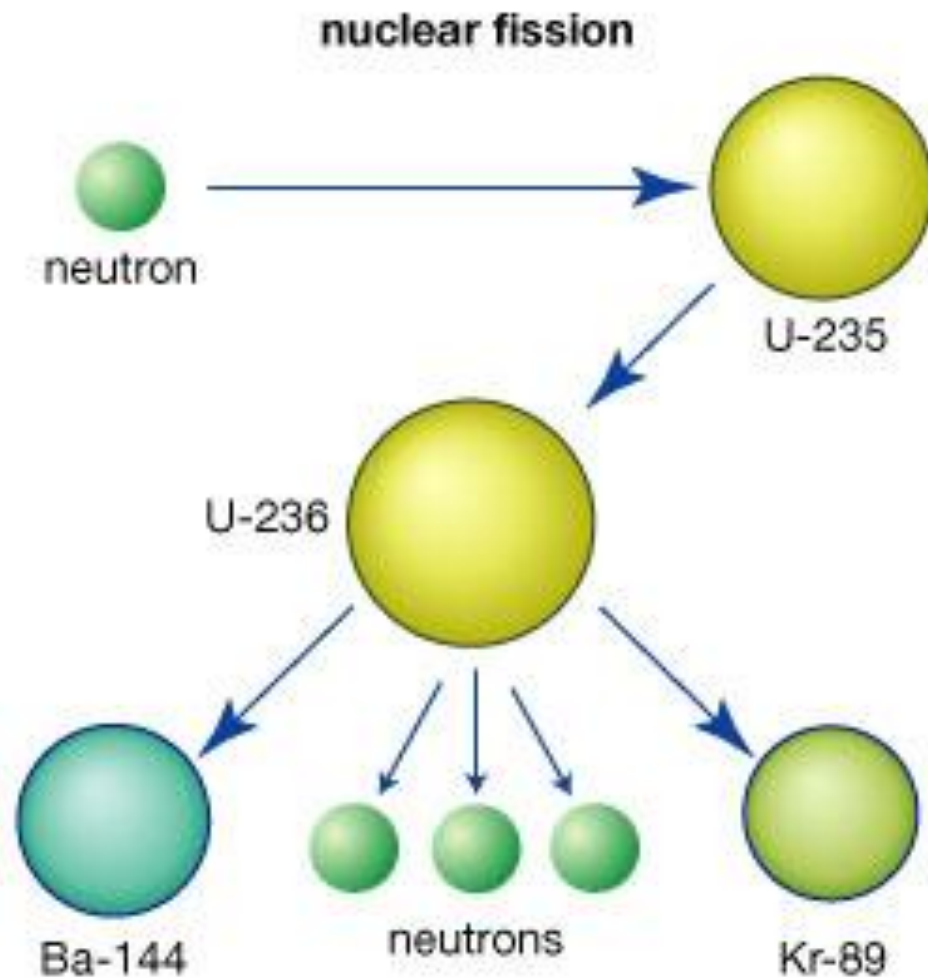
Beta Decay



Positron Decay



Fission and Fusion

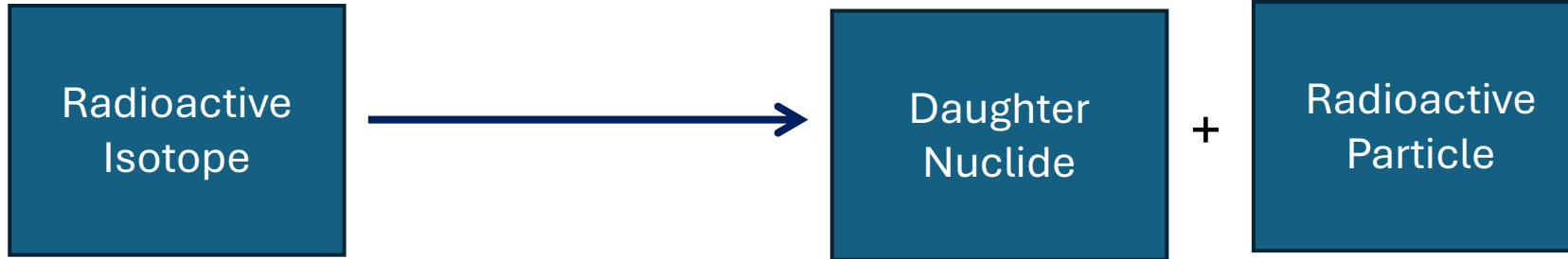


Nuclear Reactions and the Energy Released!

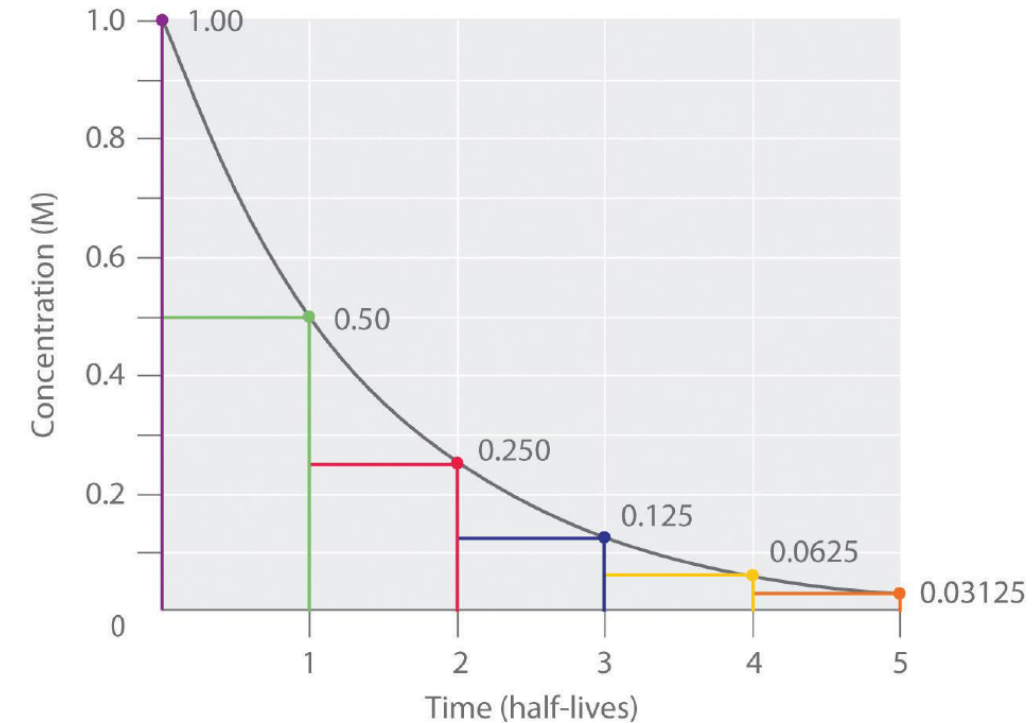
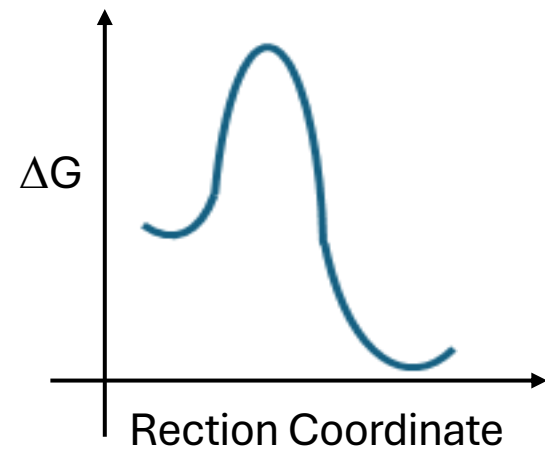


Mass Reactants	Mass Reactants	Mass Products	Mass Products
${}^{235}_{92}\text{U}$	235.04392 amu	${}^{140}_{56}\text{Ba}$	139.910581 amu
${}^1_0\text{n}$	1.00866 amu	${}^{93}_{36}\text{Kr}$	92.931130 amu
		$3 {}^1_0\text{n}$	3(1.00866) amu
Total	236.05258 amu		235.86769 amu

Curie and Nuclear Decay



Measure the time for emission of radioactive particle.



A radioactive isotope is used to indirectly monitor renal function by binding at a specific site within an enzyme. If it takes 75 minutes for the radioisotope to bind, and we need an activity of 2.4×10^{15} disintegrations per second, after binding, what mass (in grams) should be initially injected into the patient? The molar mass of the radioisotope is 114.2 grams/mole, and the half-life is 7.5 hours.

$$\text{Activity} = \frac{(\text{mass})}{(\text{Molecular Weight})} \times (\text{Avogadro's Number}) \times k \text{ (Rate Constant)}$$

Curie and Nuclear Decay

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Sodium-24 ($t_{1/2} = 15 \text{ h}$) is used to study blood circulation. If a patient is injected with an aqueous solution of $^{24}\text{NaCl}$ whose activity is $2.5 \times 10^9 \text{ d/s}$, how much of the activity is present in the patient's body and excreted in fluids after 4 days? Calculate the remaining fraction.

Learning Outcomes

- 1.) Write balanced nuclear equations for alpha decay, beta decay, and positron emission and fission processes.
- 2.) Define half-life.
- 3.) Complete kinetics calculations involving nuclear decay.

- i. Uranium-235 undergoes three decays with Actinium-227 (Ac) being the product of the third decay. Write a balanced set of nuclear reactions that depict this decay.
- ii. The halogen astatine (At) can be obtained artificially by bombarding a bismuth (Bi) target with alpha particles. It has been found to be useful for the treatment of certain types of thyroid cancers.
 - a. Write the nuclear reaction for the production of astatine from bismuth-209. Two neutrons are given off during this reaction.
 - b. One form of radioactive astatine is an alpha particle emitter with a half-life of 7.21 hrs. If a sample containing 0.100 mg of radioactive astatine is given to a person at 7 a.m. (07:00) on a given day, how much astatine will remain at 11:30 p.m. (23:30) that night?

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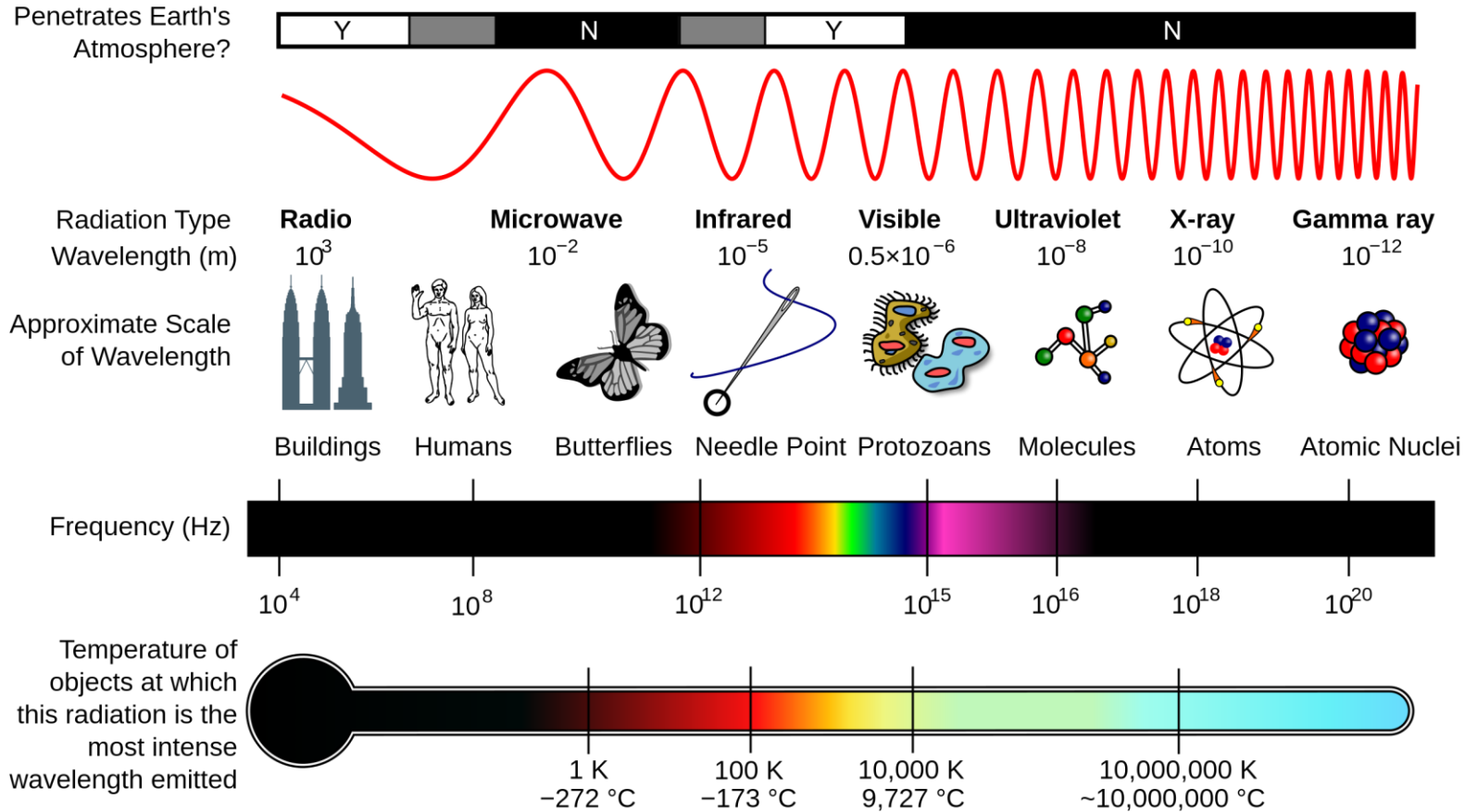
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Topics:

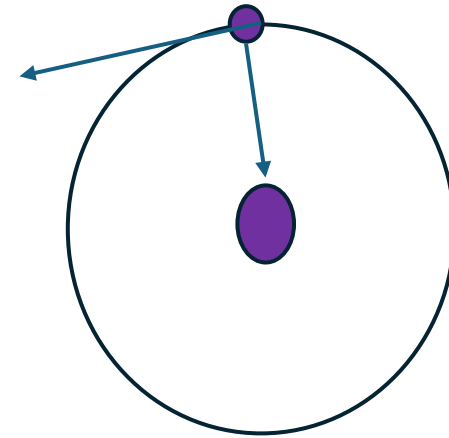
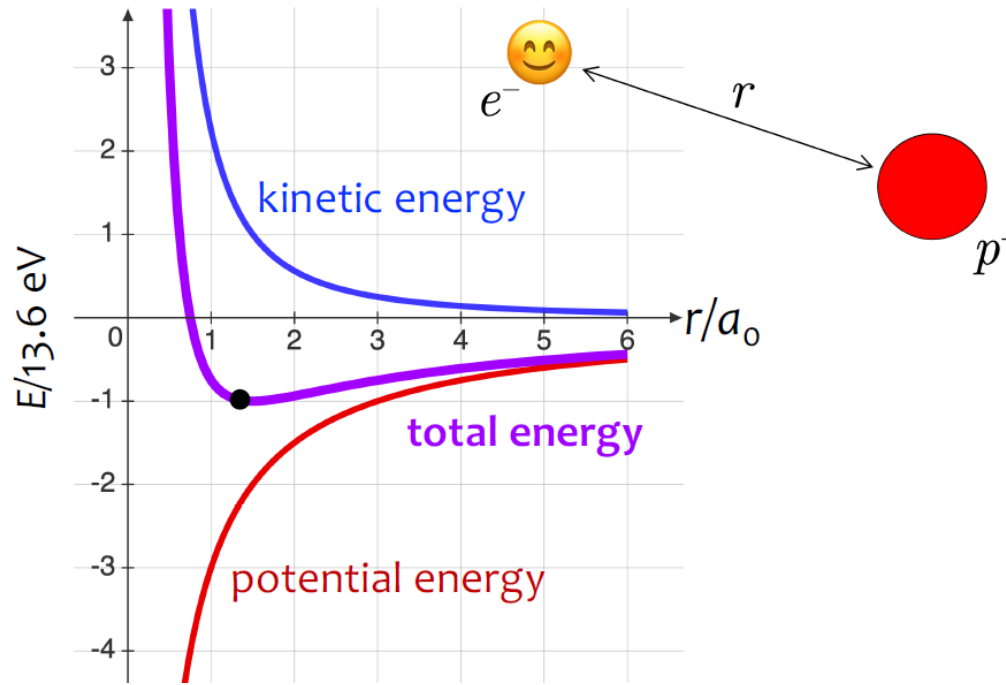
- 1.) Electromagnetic Spectrum
- 2.) The Bohr Model and Relationships with Nuclear Charge, Radii, and Velocity
- 3.) Wave-Particle Duality
- 4.) Particle-in-a-Box Model to Understand Atomic Energies
- 5.) Orbitals: Radial and Angular Nodes



From the Nuclear to Electrons



Spectroscopy & the Bohr Model

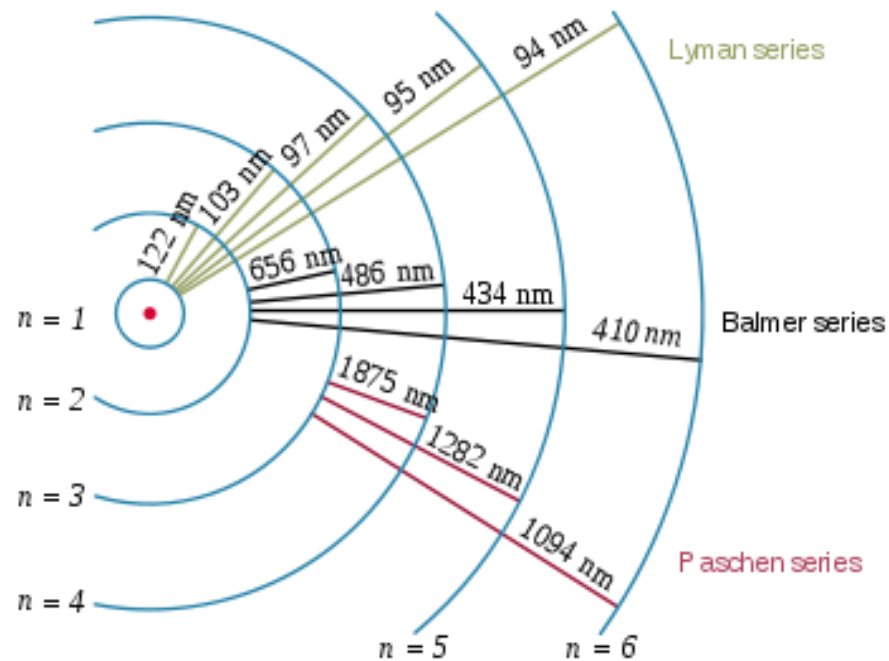
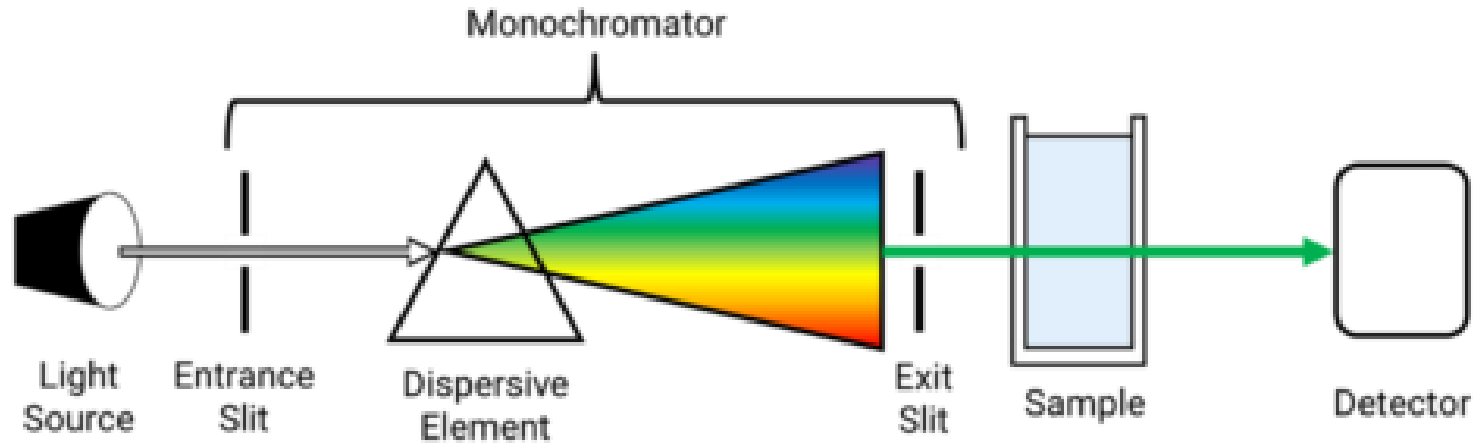


$$mvr = \frac{nh}{2\pi}$$

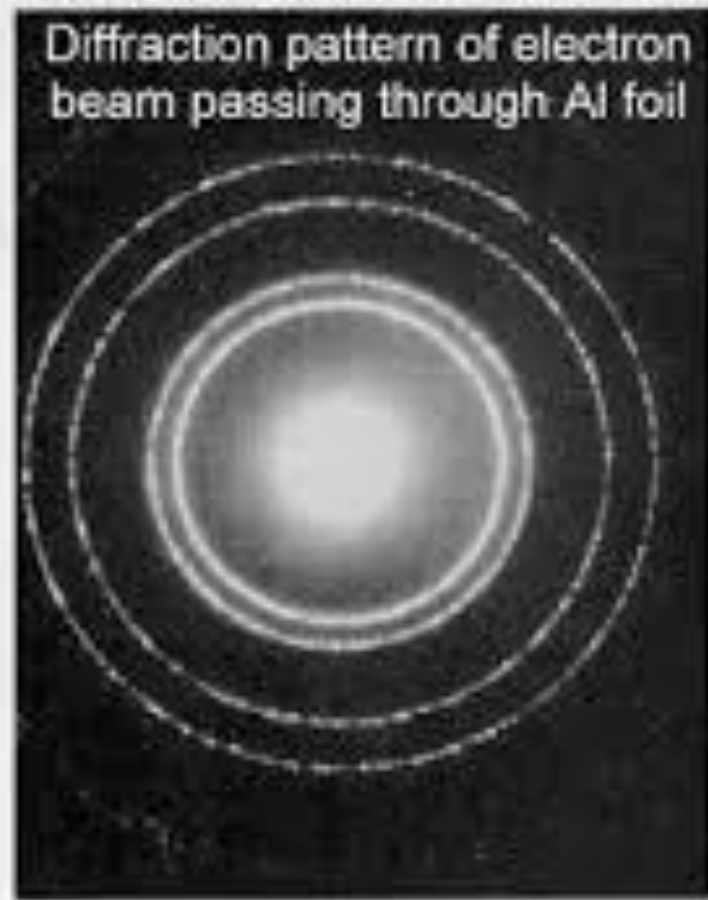
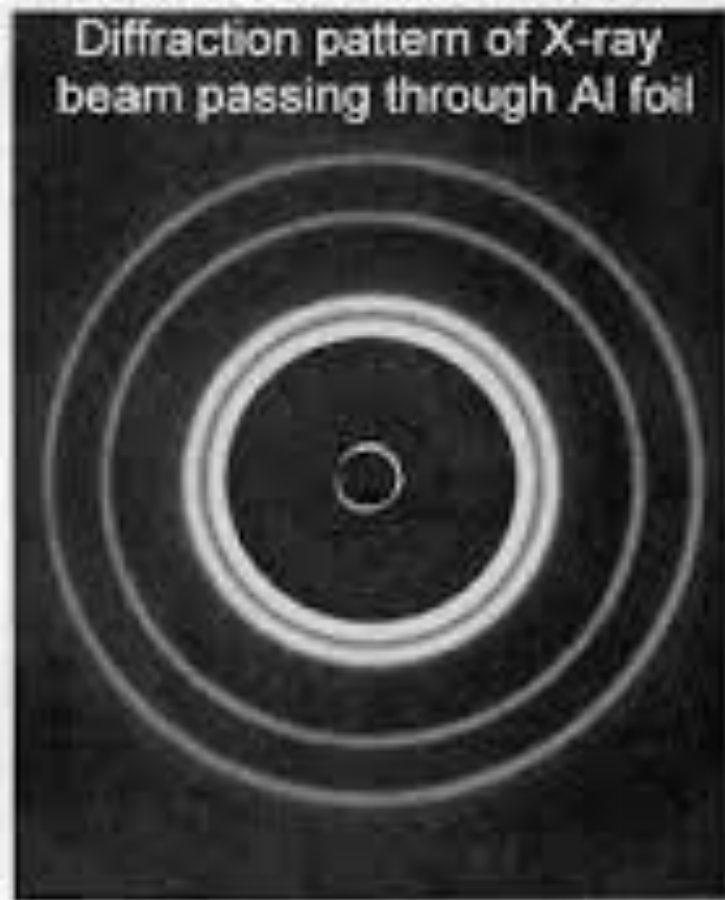
$$F_{\text{centrifugal}} = \frac{mv^2}{r}$$

$$F_{\text{centripetal}} = \frac{Ze^2}{kr^2}$$

Spectroscopy & the Bohr Model



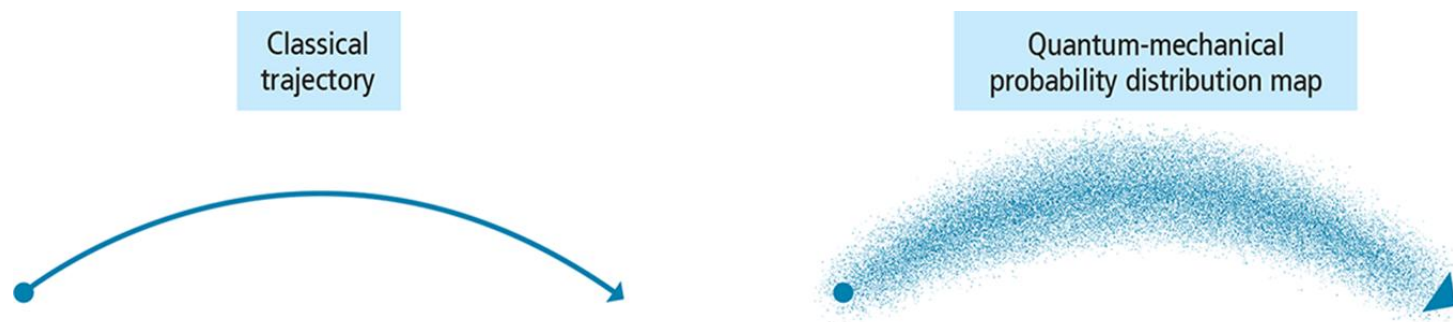
The Failure of the Bohr Model



$$\text{de Broglie Wavelength} = \frac{h}{mv}$$

Determinacy versus Indeterminacy

- According to classical physics, particles move in a path **determined** by the particle's velocity, position, and forces acting on it.
 - **Determinacy** = definite, predictable future
- Because we cannot know both the position and velocity of an electron, we cannot predict the path it will follow.
 - **Indeterminacy** = indefinite future, can predict only probability
- The best we can do is to describe the probability an electron will be found in a particular region using statistical functions.

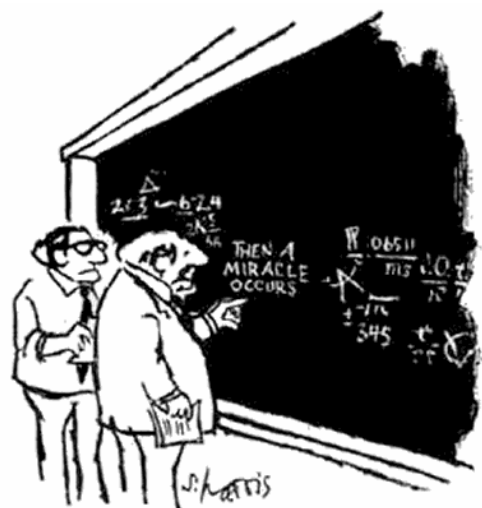


$$\hat{H}\psi = E\psi$$

- What is the basis for the equation? What are the assumptions?
- The solution produced **wave functions**, ψ , which are mathematical functions that describe the wavelike nature of the electron. What are the key components in wave functions that provide information about electrons?

Schrödinger's Equation

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"I THINK YOU SHOULD BE MORE EXPLICIT
HERE IN STEP TWO."

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Solving the Equation

$$-\frac{h^2}{4\pi^2m} \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \right) \Psi = E\Psi$$

Classical
trajectory

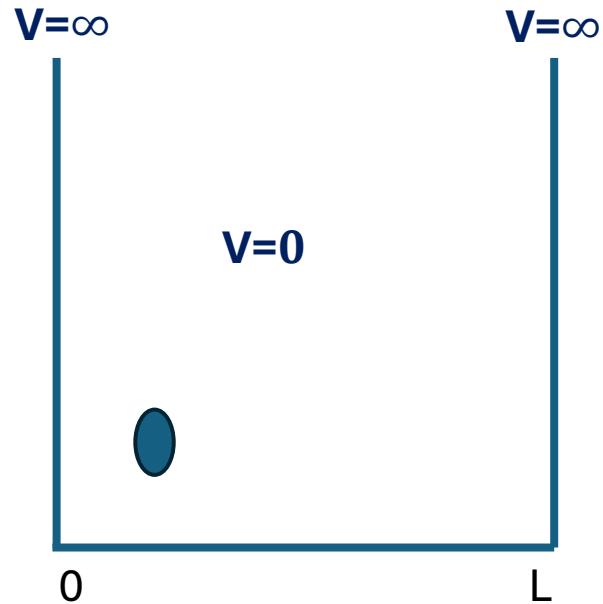


Quantum-mechanical
probability distribution map

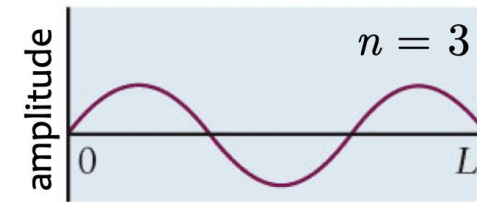


1D Particle-in-a-Box

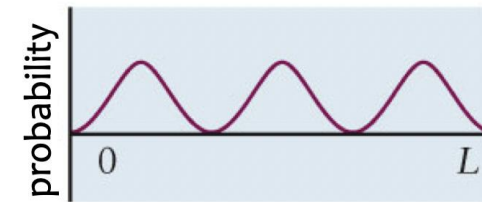
A simplistic model that allows us to get information about electronic behavior.



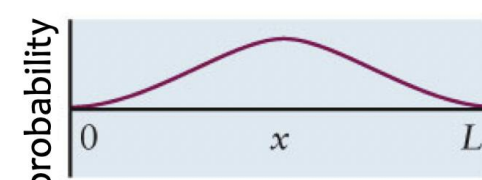
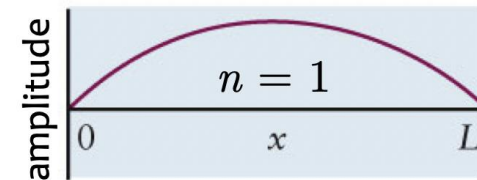
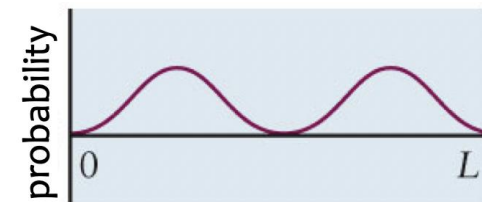
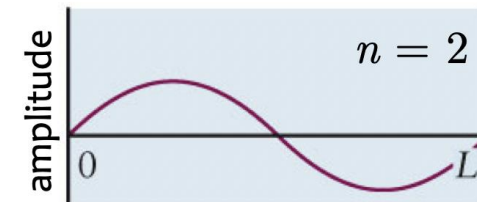
$$-\frac{h^2}{8\pi^2m} \left(\frac{d^2}{dx^2} \right) \Psi = E\Psi$$



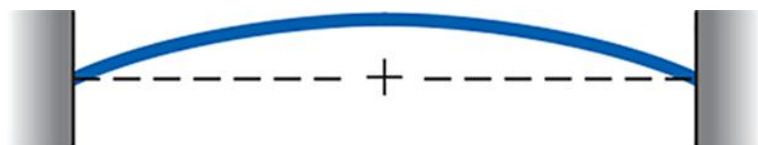
wavefunction (ψ)



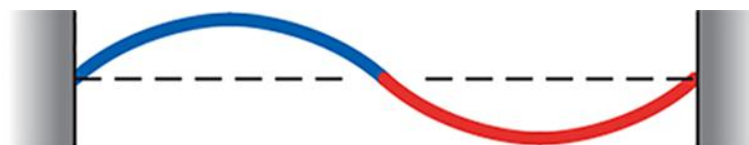
probability (ψ^2)



Angular Component



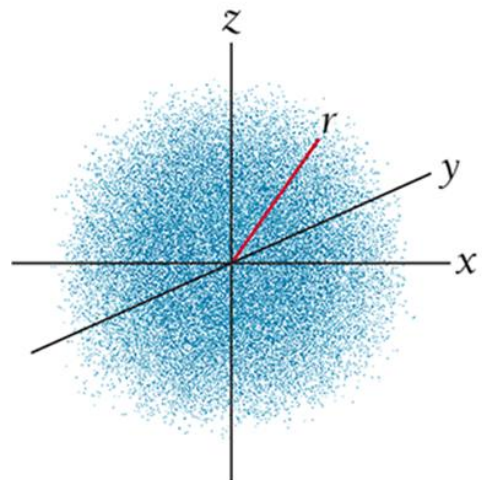
1s orbital



2p orbital

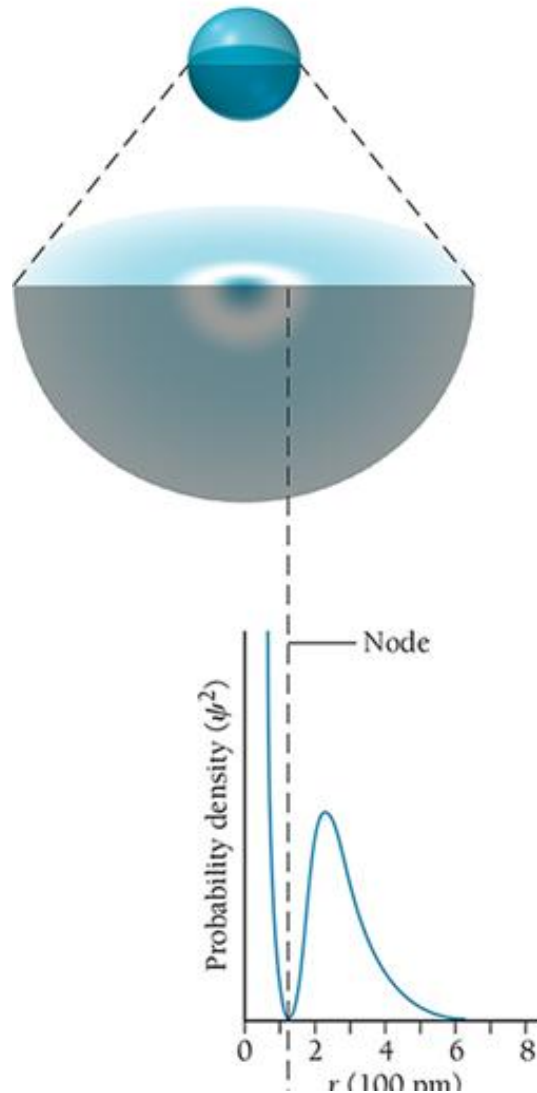
s-orbitals and Radial Distribution

1s orbital

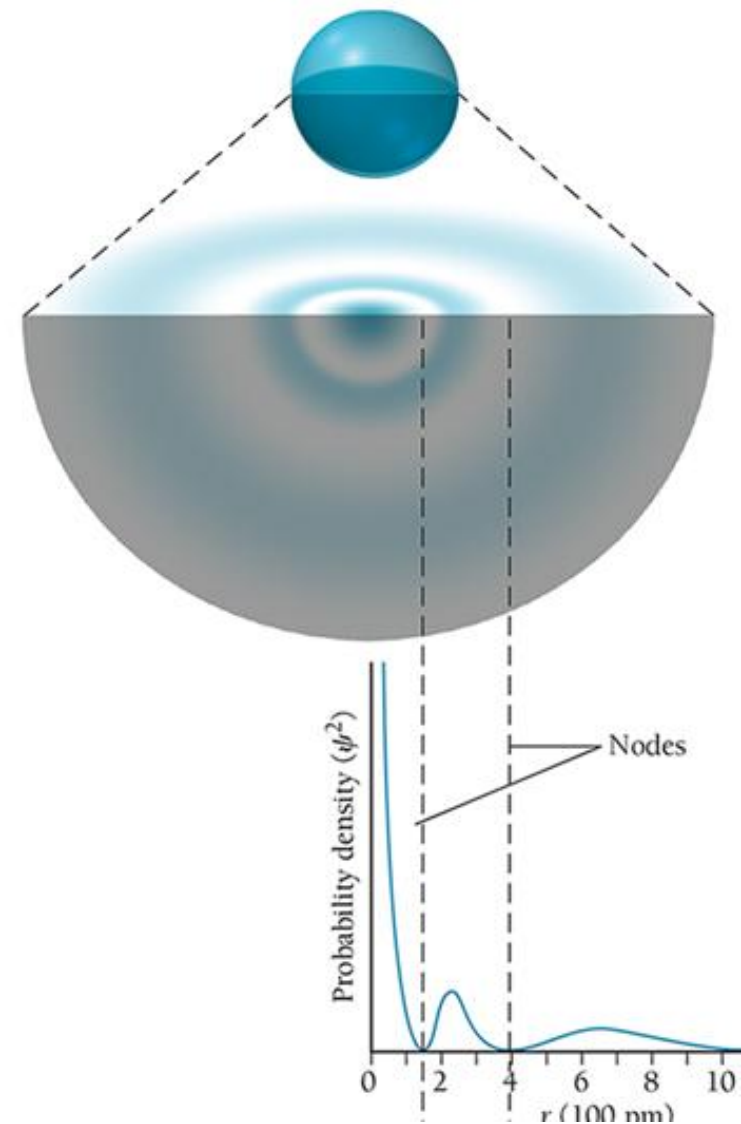


Density of dots
proportional to
probability density (ψ^2).

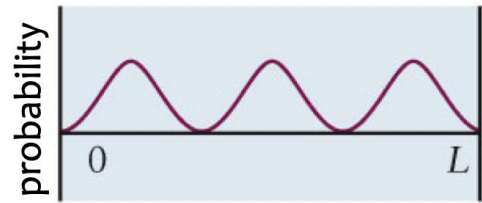
2s ($n = 2, l = 0$)



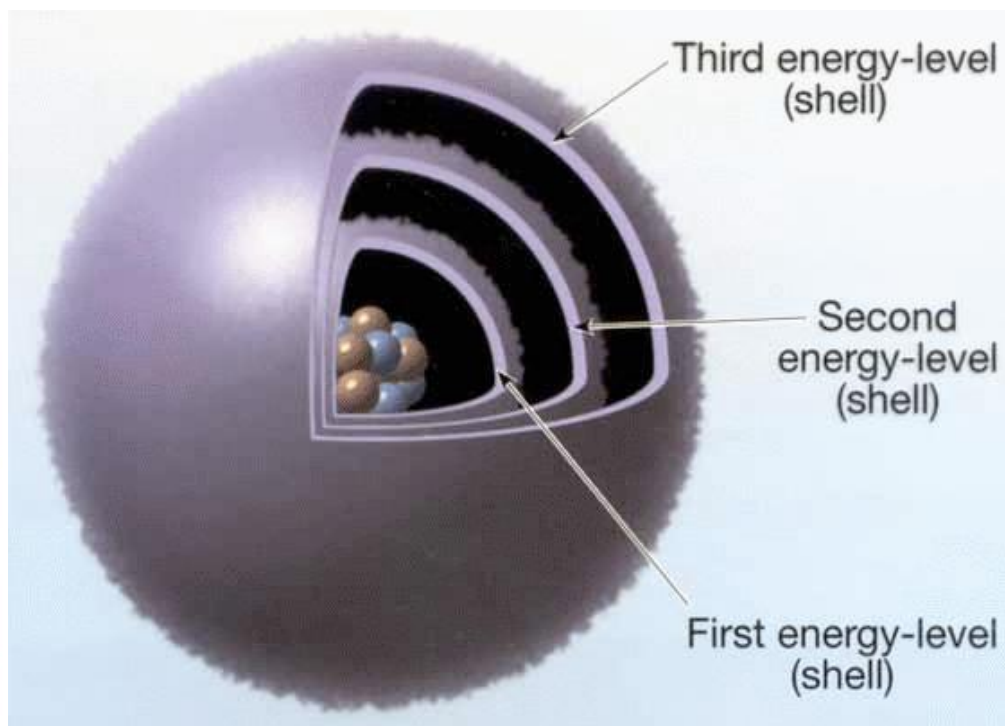
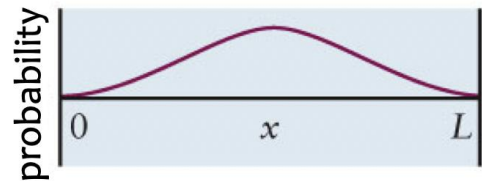
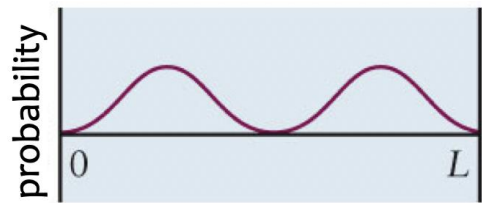
3s ($n = 3, l = 0$)



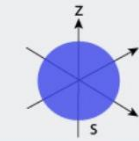
Orbitals and Quantum Numbers



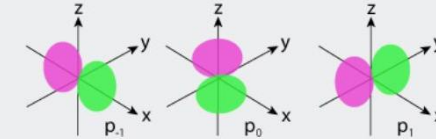
probability (ψ^2)



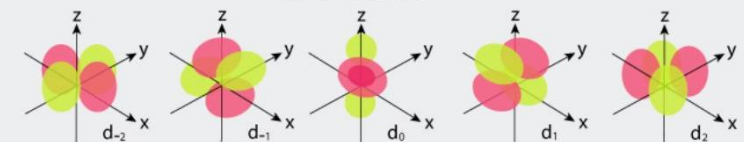
1. s-orbital



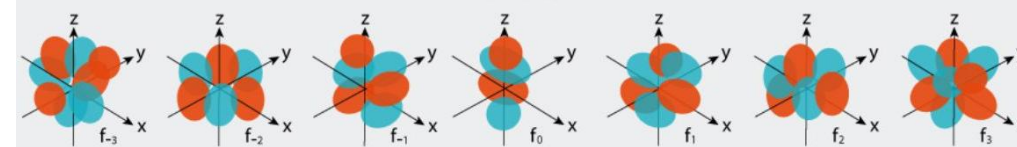
2. p-orbital



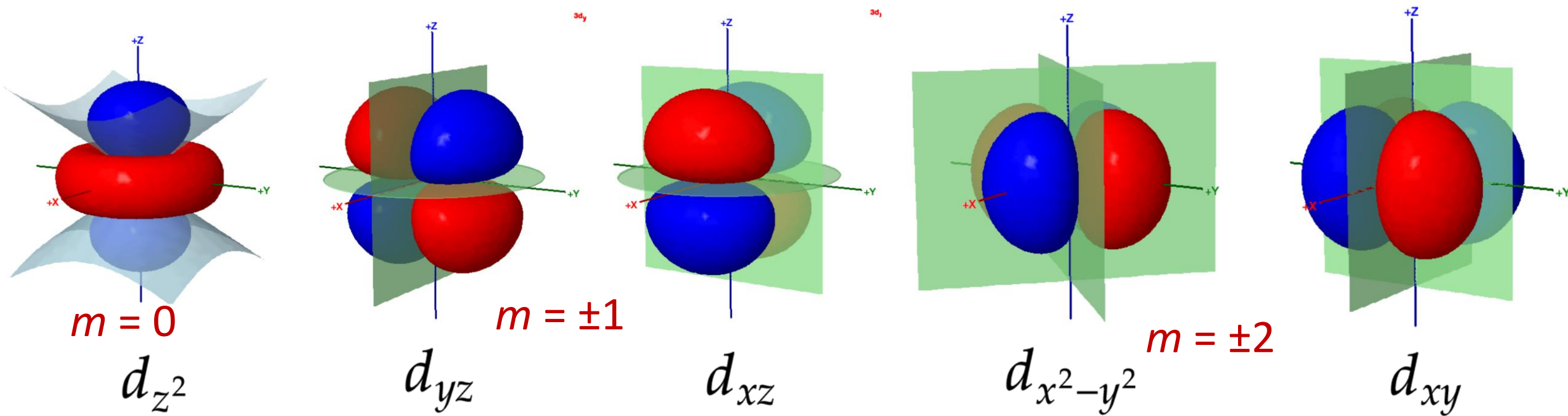
3. d-orbital



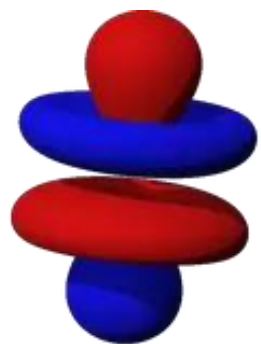
4. f-orbital



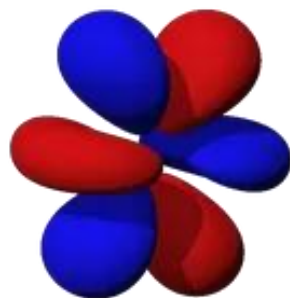
Angular Component



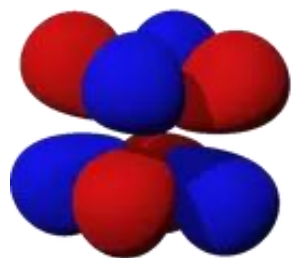
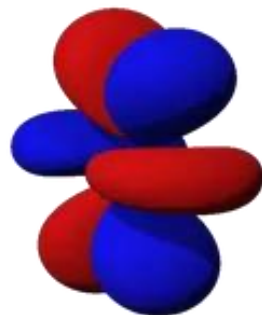
Angular Component



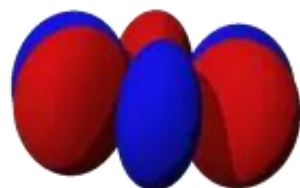
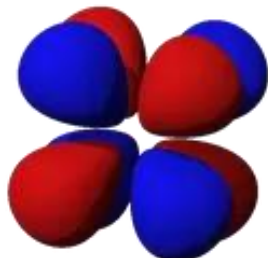
$m = 0$



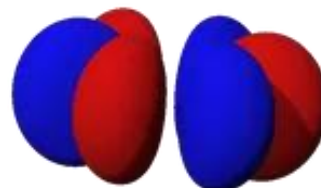
$m = \pm 1$



$m = \pm 2$

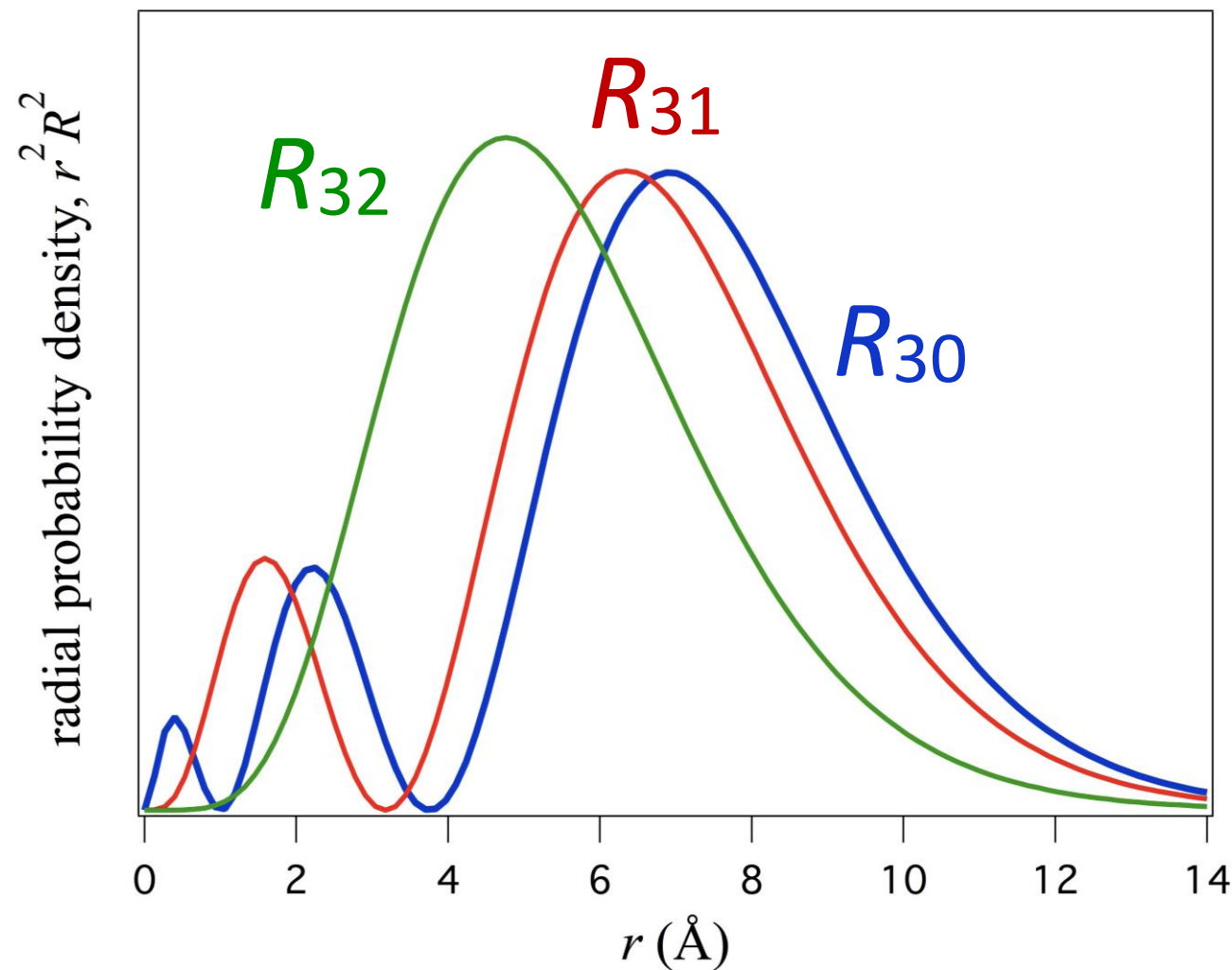


$m = \pm 3$

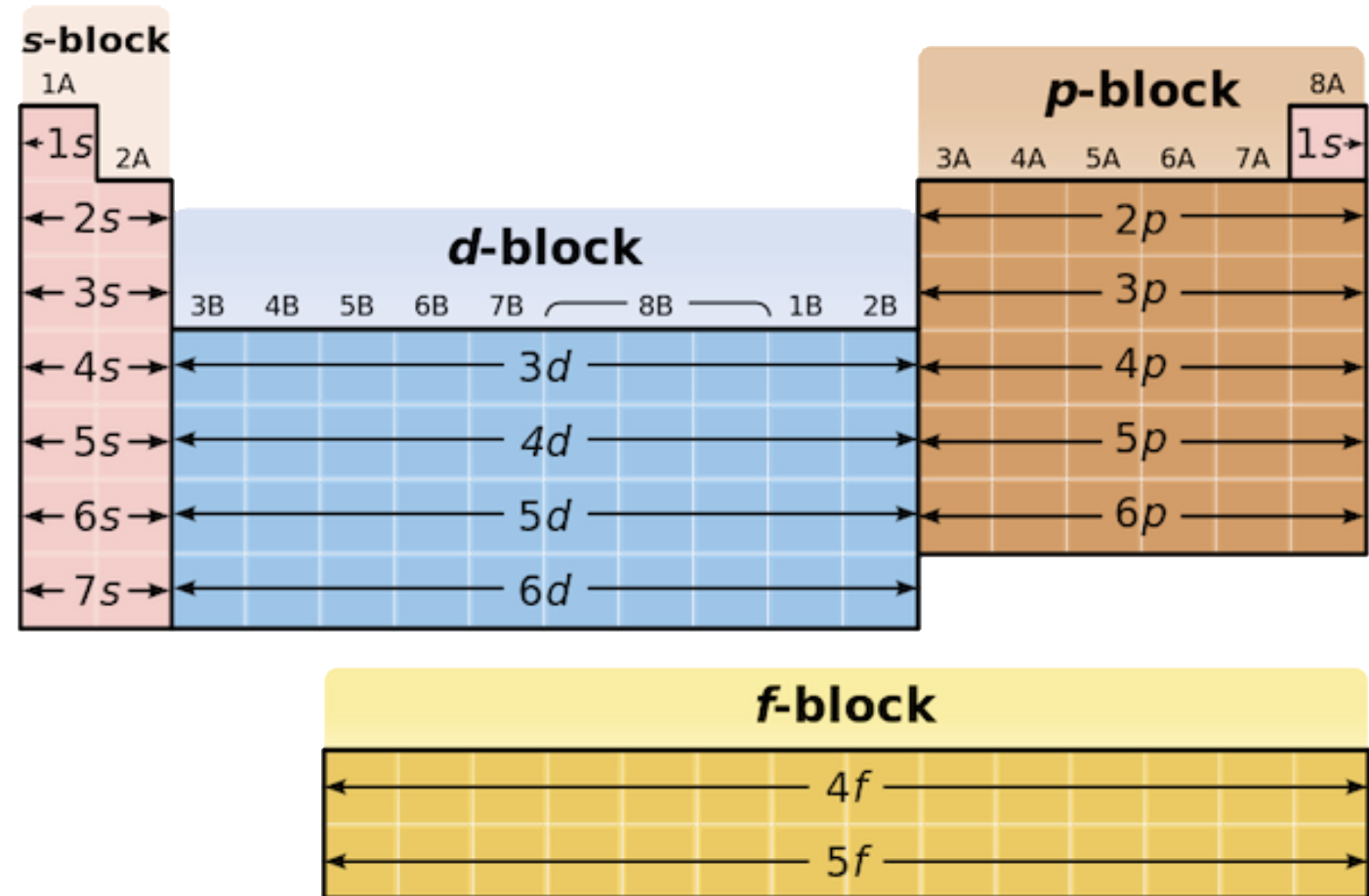
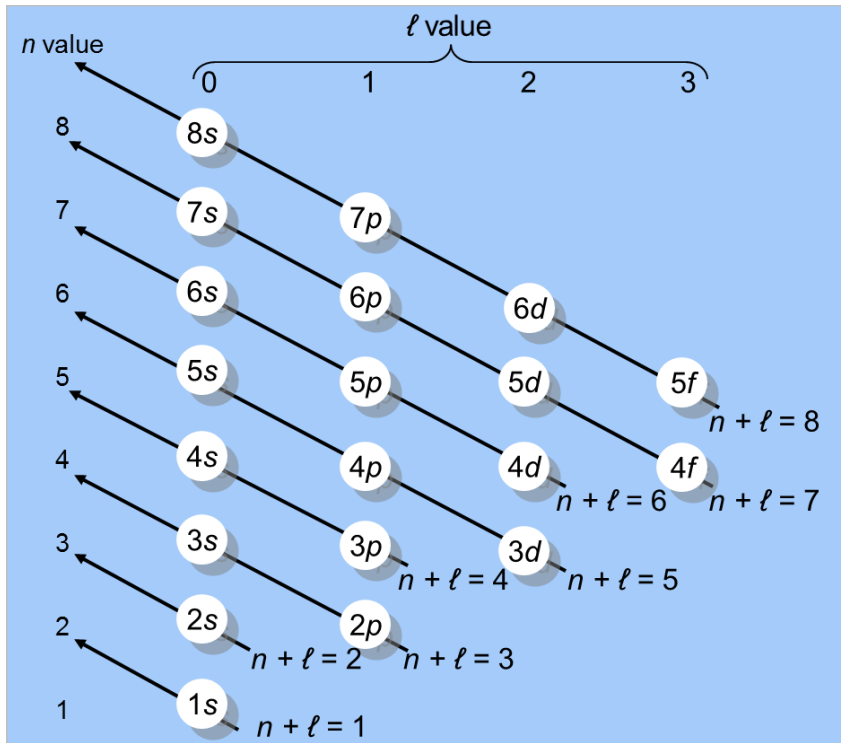


3s, 3p, and 3d Radial Comparison

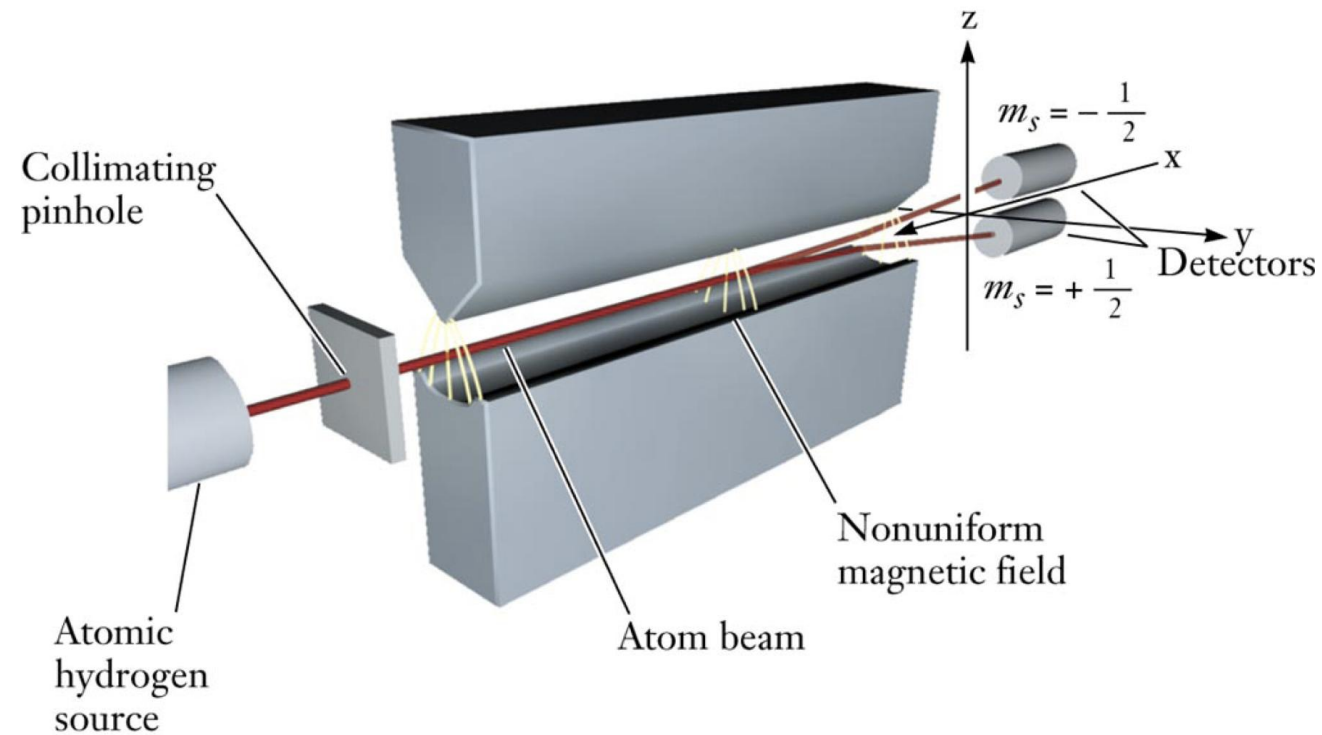
For a given n , the most probable distance from the nucleus at which we expect to find an e^- decreases with increasing ℓ .



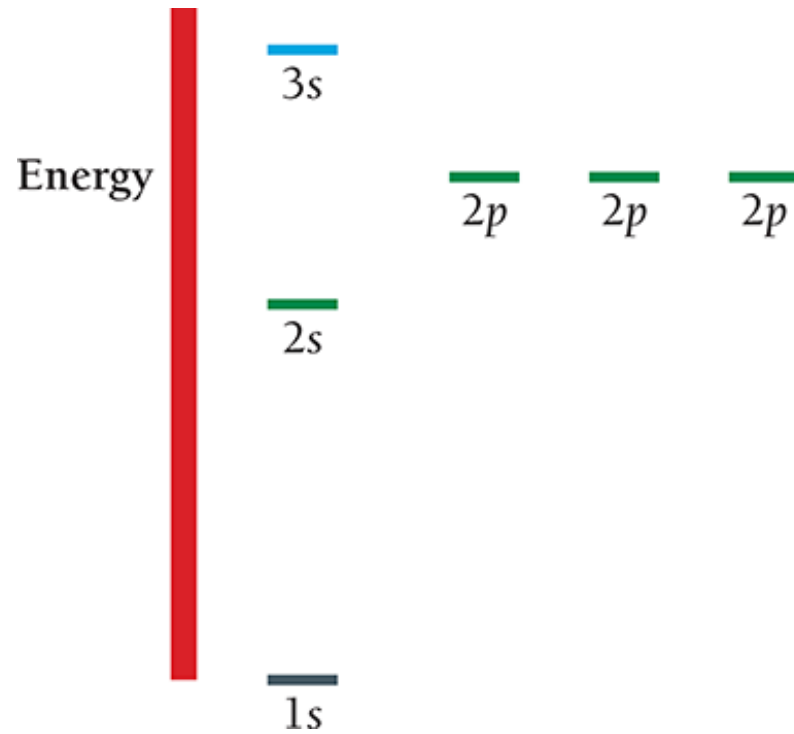
Electron Configurations



The Magnetic Quantum Number



Atomic Orbital Organization



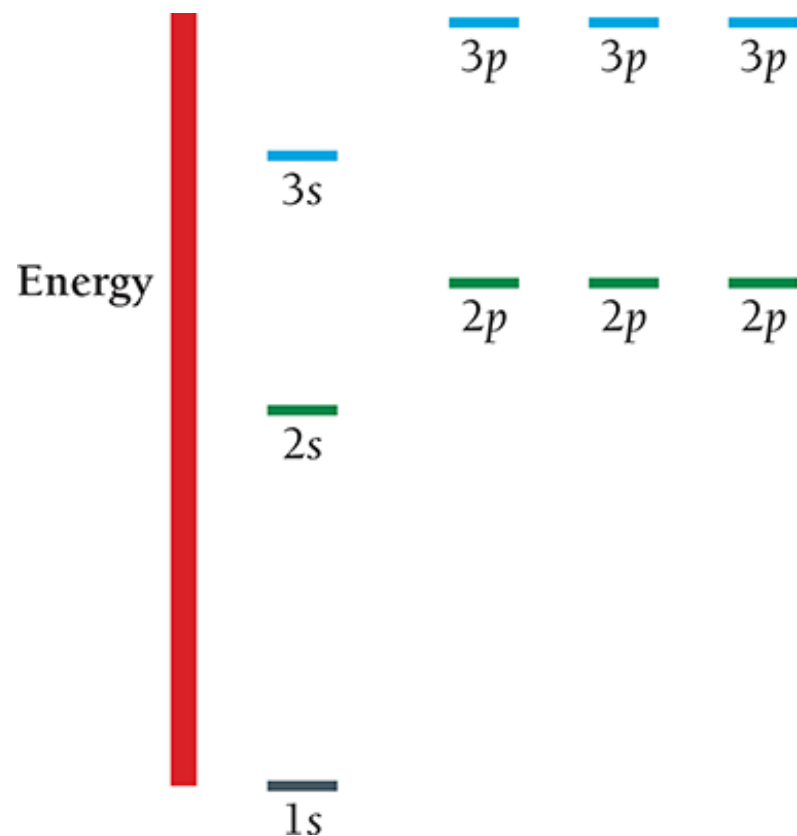
Atomic Symbol – Neon, Ne

Define the following:

Aufbau

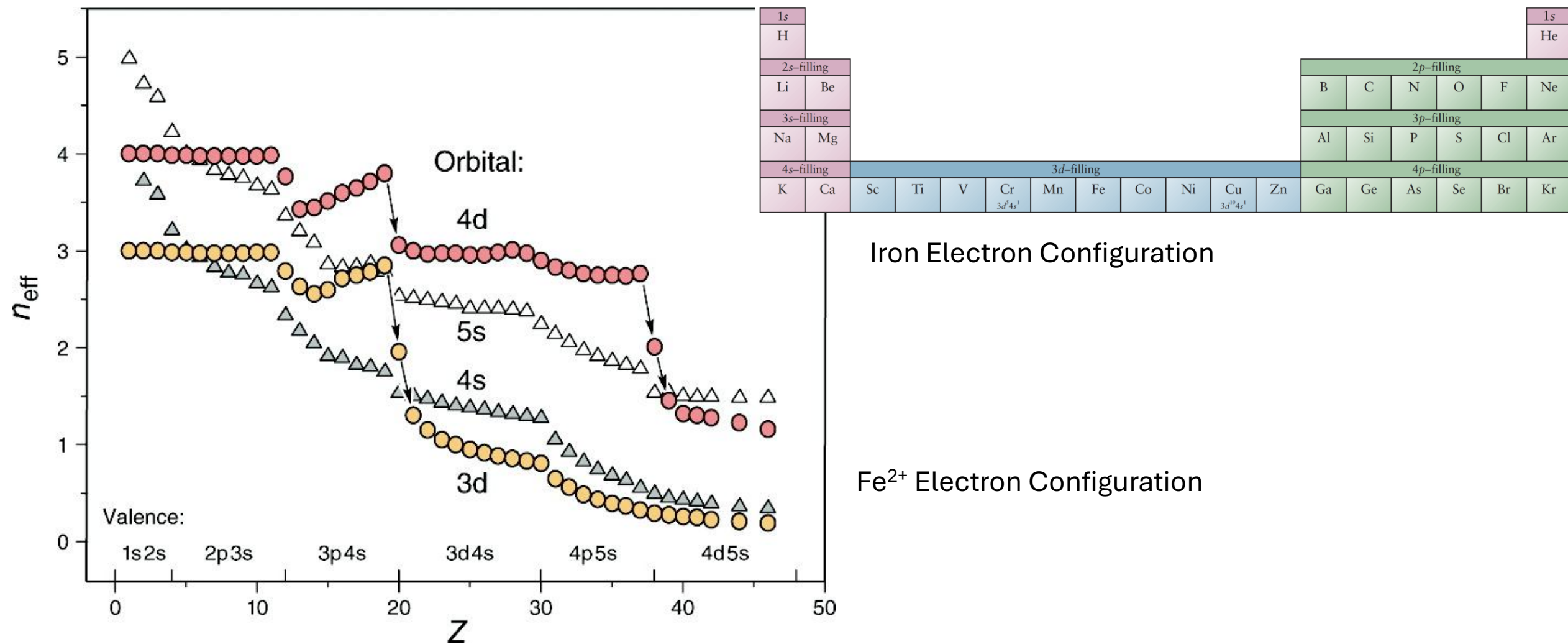
Hund's Rule

Pauli Exclusion



The electrons in all the sublevels with the highest principal energy shell are called the **valence electrons**.
Electrons in lower energy shells are called **core electrons**.
One of the most important factors in the way an atom behaves, both chemically and physically, is the number of valence electrons.

Duke University - Chem 210
Prof. Cox



Aufbau Doesn't Always HOLD!

Element Name and Symbol	Atomic Number	Common Oxidation States	Electron Configuration	
Scandium (Sc)	21	+3	Sc: [Ar] 4s ² 3d ¹	Sc: [Ar] $\frac{1\downarrow}{4s}$ $\frac{1}{\underbrace{\quad\quad\quad}_{3d}}$
Titanium (Ti)	22	+4	Ti: [Ar] 4s ² 3d ²	Ti: [Ar] $\frac{1\downarrow}{4s}$ $\frac{1}{\underbrace{\quad\quad}_{3d}}$ $\frac{1}{\quad}$
Vanadium (V)	23	+2, +3, +4, +5	V: [Ar] 4s ² 3d ³	V: [Ar] $\frac{1\downarrow}{4s}$ $\frac{1}{\underbrace{\quad\quad}_{3d}}$ $\frac{1}{\quad}$ $\frac{1}{\quad}$
Chromium (Cr)	24	+2, +3, +6	Cr: [Ar] 4s ¹ 3d ⁵	Cr: [Ar] $\frac{1}{4s}$ $\frac{1}{\underbrace{\quad\quad}_{3d}}$ $\frac{1}{\quad}$ $\frac{1}{\quad}$ $\frac{1}{\quad}$ $\frac{1}{\quad}$
Manganese (Mn)	25	+2, +3, +4, +6, +7	Mn: [Ar] 4s ² 3d ⁵	Mn: [Ar] $\frac{1\downarrow}{4s}$ $\frac{1}{\underbrace{\quad\quad}_{3d}}$ $\frac{1}{\quad}$ $\frac{1}{\quad}$ $\frac{1}{\quad}$ $\frac{1}{\quad}$
Iron (Fe)	26	+2, +3	Fe: [Ar] 4s ² 3d ⁶	Fe: [Ar] $\frac{1\downarrow}{4s}$ $\frac{1\downarrow}{\underbrace{\quad\quad}_{3d}}$ $\frac{1}{\quad}$ $\frac{1}{\quad}$ $\frac{1}{\quad}$ $\frac{1}{\quad}$
Cobalt (Co)	27	+2, +3	Co: [Ar] 4s ² 3d ⁷	Co: [Ar] $\frac{1\downarrow}{4s}$ $\frac{1\downarrow}{\underbrace{\quad\quad}_{3d}}$ $\frac{1\downarrow}{\quad}$ $\frac{1}{\quad}$ $\frac{1}{\quad}$ $\frac{1}{\quad}$
Nickel (Ni)	28	+2	Ni: [Ar] 4s ² 3d ⁸	Ni: [Ar] $\frac{1\downarrow}{4s}$ $\frac{1\downarrow}{\underbrace{\quad\quad}_{3d}}$ $\frac{1\downarrow}{\quad}$ $\frac{1\downarrow}{\quad}$ $\frac{1}{\quad}$ $\frac{1}{\quad}$
Copper (Cu)	29	+1, +2	Cu: [Ar] 4s ¹ 3d ¹⁰	Cu: [Ar] $\frac{1}{4s}$ $\frac{1\downarrow}{\underbrace{\quad\quad}_{3d}}$ $\frac{1\downarrow}{\quad}$ $\frac{1\downarrow}{\quad}$ $\frac{1\downarrow}{\quad}$ $\frac{1\downarrow}{\quad}$
Zinc (Zn)	30	+2	Zn: [Ar] 4s ² 3d ¹⁰	Zn: [Ar] $\frac{1\downarrow}{4s}$ $\frac{1\downarrow}{\underbrace{\quad\quad}_{3d}}$ $\frac{1\downarrow}{\quad}$ $\frac{1\downarrow}{\quad}$ $\frac{1\downarrow}{\quad}$ $\frac{1\downarrow}{\quad}$

Learning Outcomes

- 1.) Explain the relationships that are deduced between kinetic and potential energies from the Bohr Model.
- 2.) Describe the relationship between energies and the number of nodes.
- 3.) Write electron configurations for elements through Kr.
- 4.) Identify the quantum numbers associated with specific electrons in an atom.

i. Write the electron configurations for Co, Co^{2+} , and Co^{3+} .

ii. What are the centripetal and centrifugal forces in the Bohr model? How are these forces related (which is greater or are they equal)?

iii. The “g” orbitals do exist! They come after the f-orbitals and are observed in excited states for atoms. How many radial and angular nodes are in a 9g orbital?

CHEM 210

Modern Applications of Chemistry

Duke University

Prof. Charlie Cox

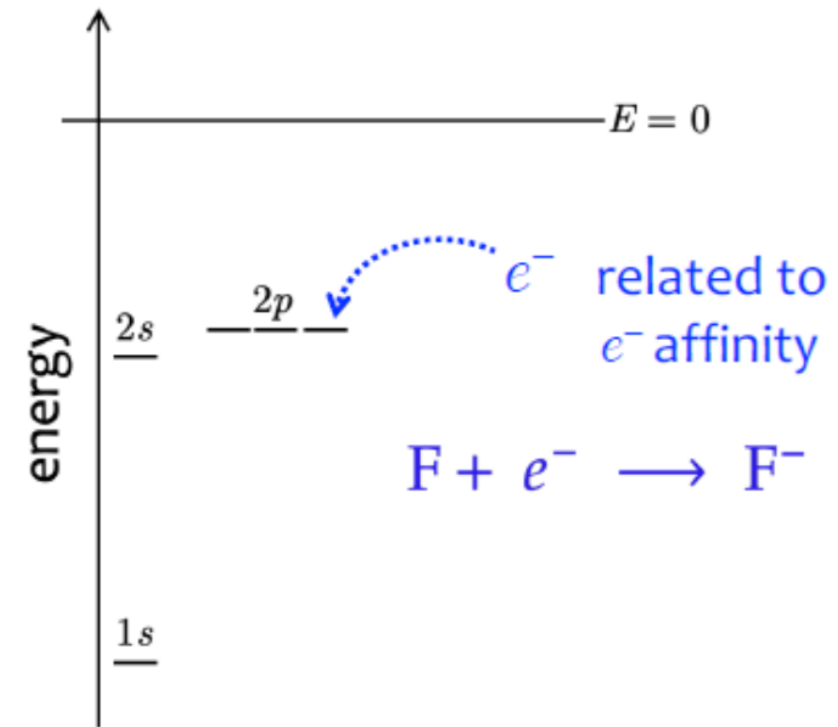
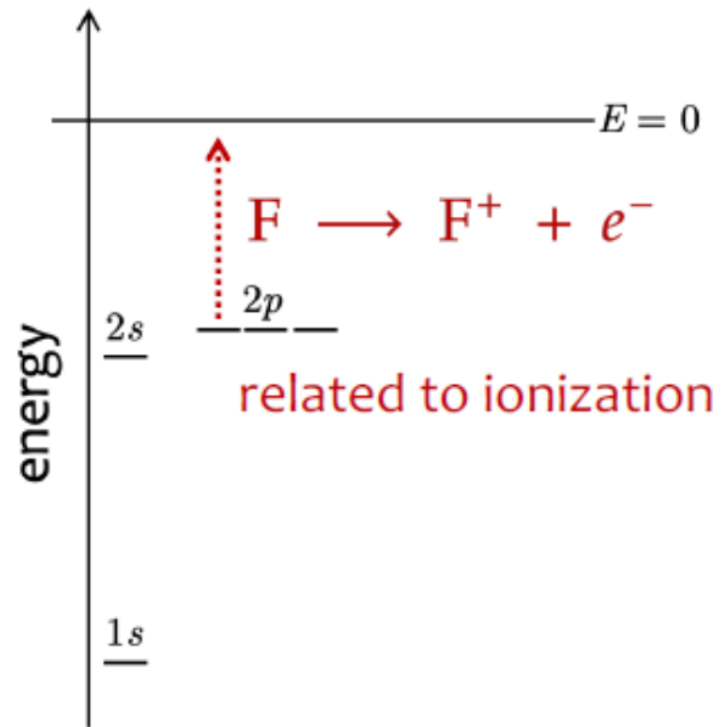
Topics:

- 1.) Ionization Energies
- 2.) Effective Nuclear Charge
- 3.) Atomic Radii
- 4.) The Lanthanide Contraction
- 5.) Electronegativity Scales
- 6.) Relativistic Effects

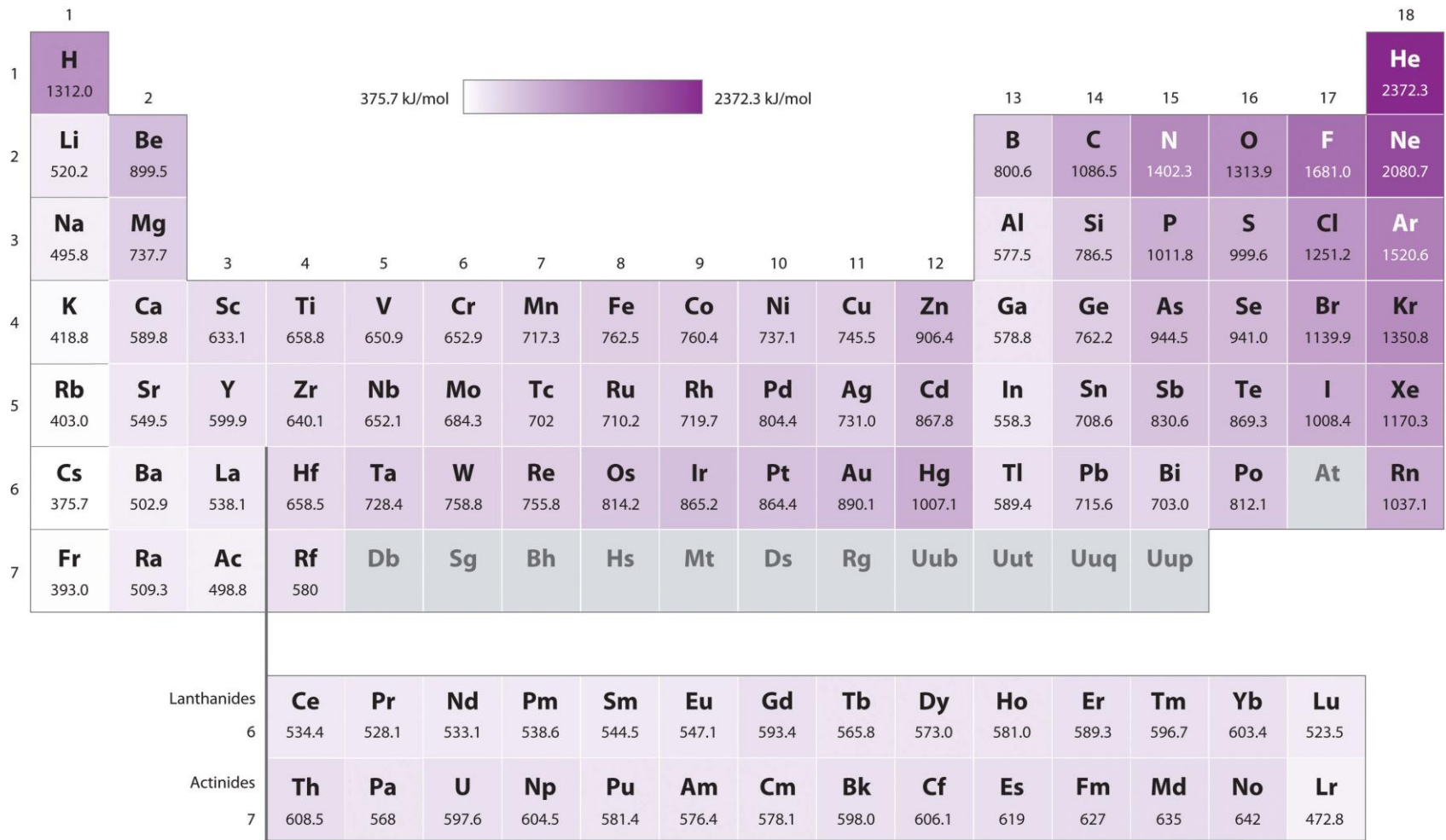


Electron Removal and Addition

Ionization energy and electron affinity are related to valence orbital energies.

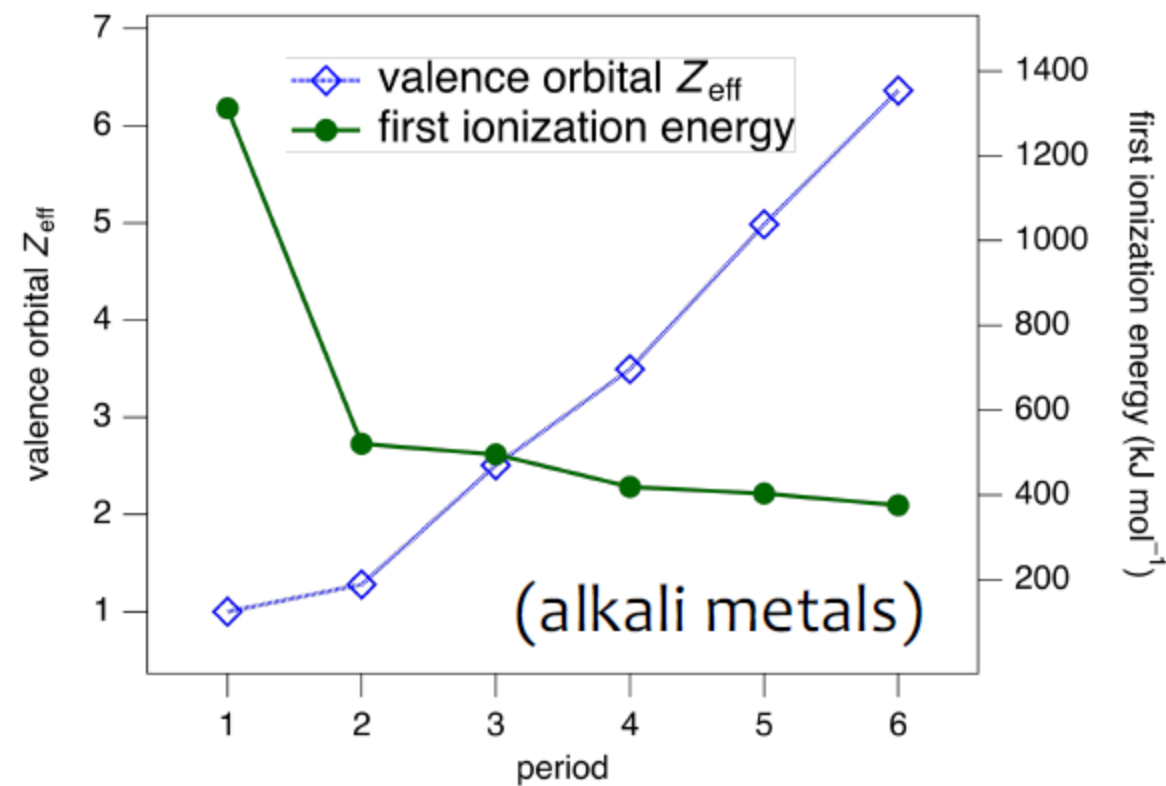
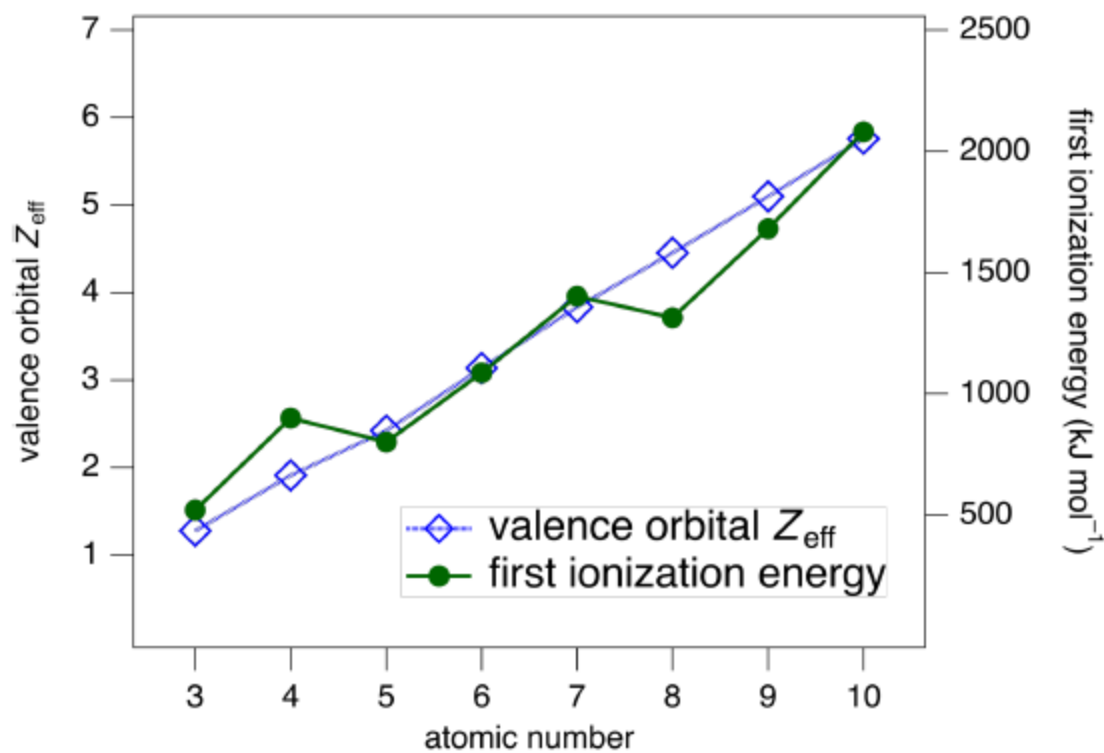


Ionization Energies



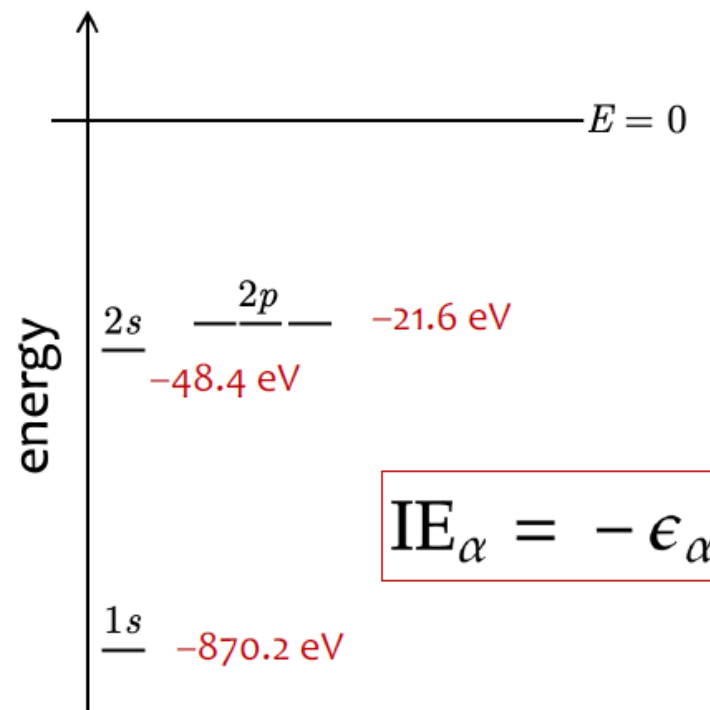
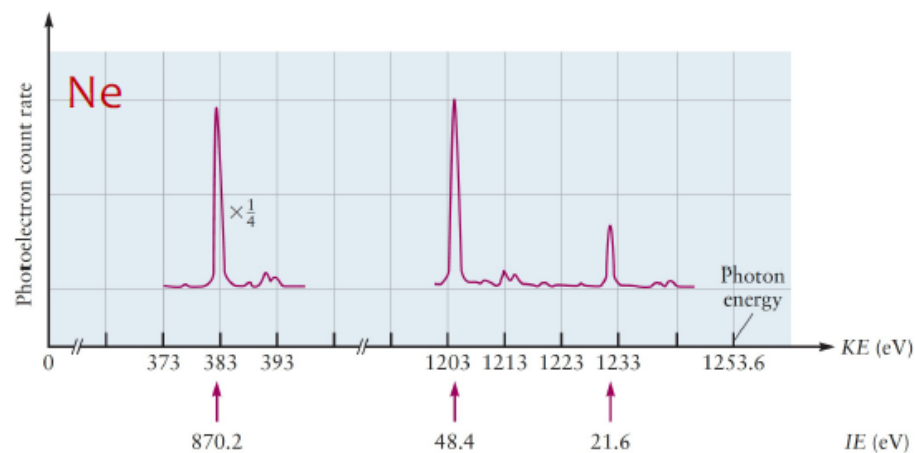
Trends in Ionization Energies

Ionization energy is related to Z_{eff} when traversing a period.



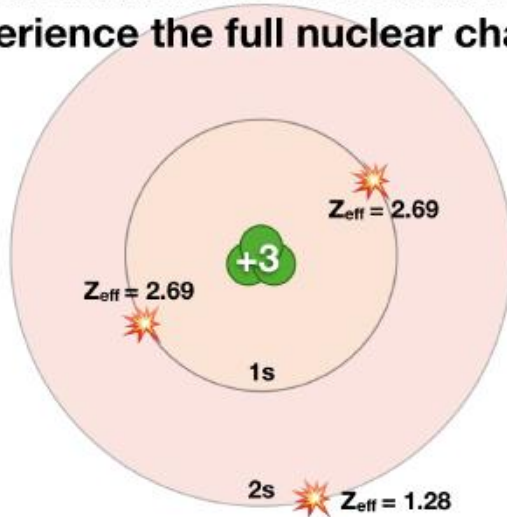
Photoelectron Spectroscopy

Orbital energies can be measured using *photoelectron spectroscopy*.



Slater Rules and Shielding

Electrons in multi electron atoms **do not** experience the full nuclear charge!



Effective nuclear charge (Z_{eff}) is the charge that each electron experiences.

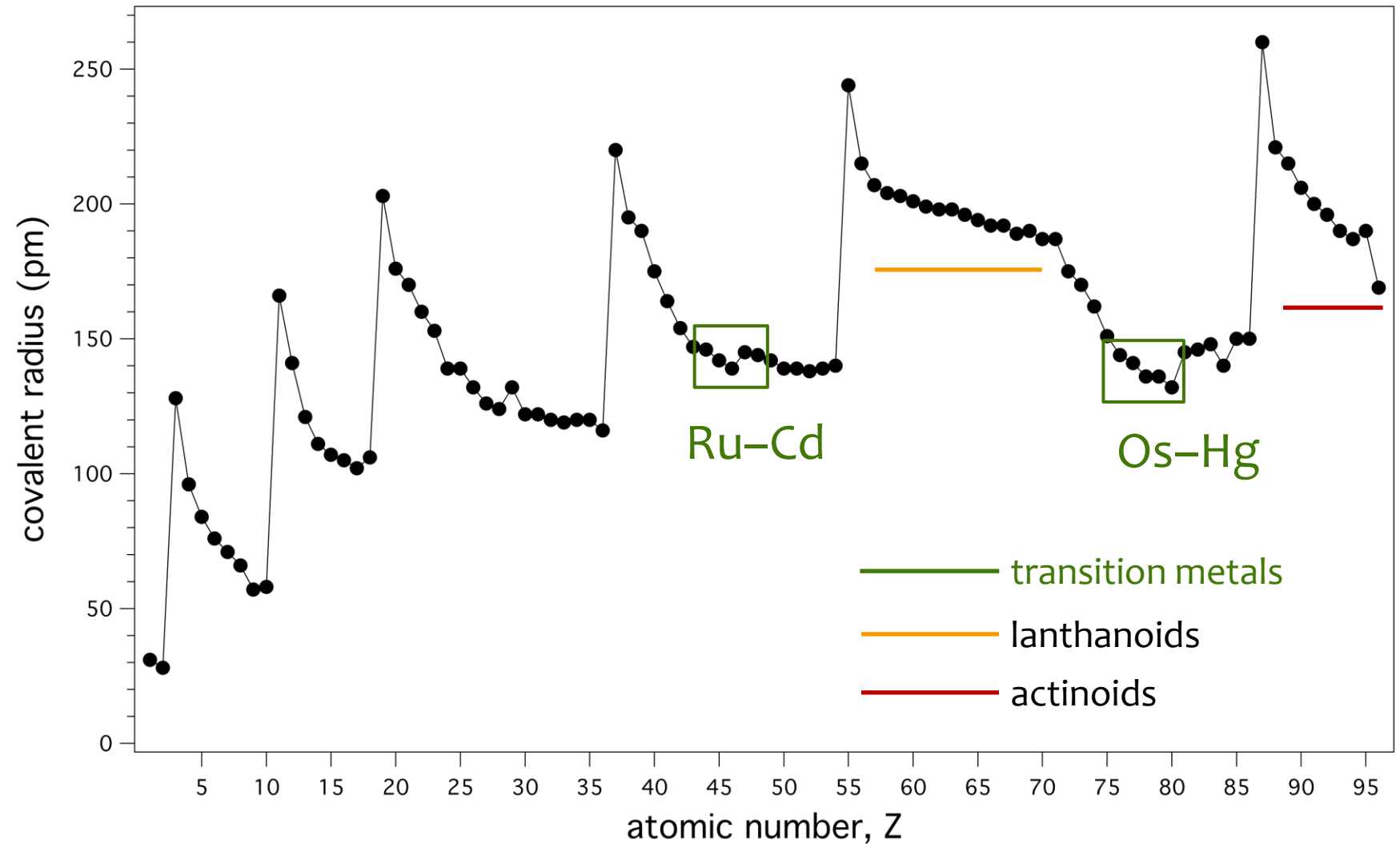
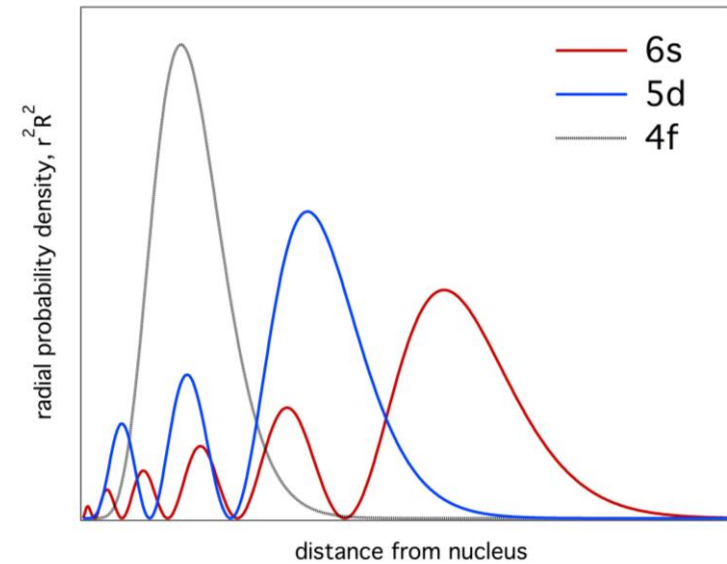
- 1.) Electrons within the same shell, shield other electrons in that shell by 35%.
- 2.) Electrons in an $n-1$ shell shield the outer electrons by 85%.
- 3.) Electrons in an $n-2$ shell and beyond shield the outer electrons by 100%.

ATOMIC RADIUS (pm)

H 38																	He 32
Li 134	Be 90											B 82	C 77	N 75	O 73	F 71	Ne 69
Na 154	Mg 130											Al 118	Si 111	P 106	S 102	Cl 99	Ar 97
K 196	Ca 174	Sc 144	Ti 136	V 125	Cr 127	Mn 139	Fe 125	Co 126	Ni 121	Cu 138	Zn 131	Ga 126	Ge 122	As 119	Se 116	Br 114	Kr 110
Rb 211	Sr 192	Y 162	Zr 148	Nb 137	Mo 145	Tc 156	Ru 126	Rh 135	Pd 131	Ag 153	Cd 148	In 144	Sn 141	Sb 138	Te 135	I 133	Xe 130
Cs 225	Ba 198		Hf 150	Ta 138	W 146	Re 159	Os 128	Ir 137	Pt 128	Au 144	Hg 149	Tl 148	Pb 147	Bi 146	Po	At	Rn 145

Lanthanide Contraction

A complete 4f subshell effects the subsequent elements.

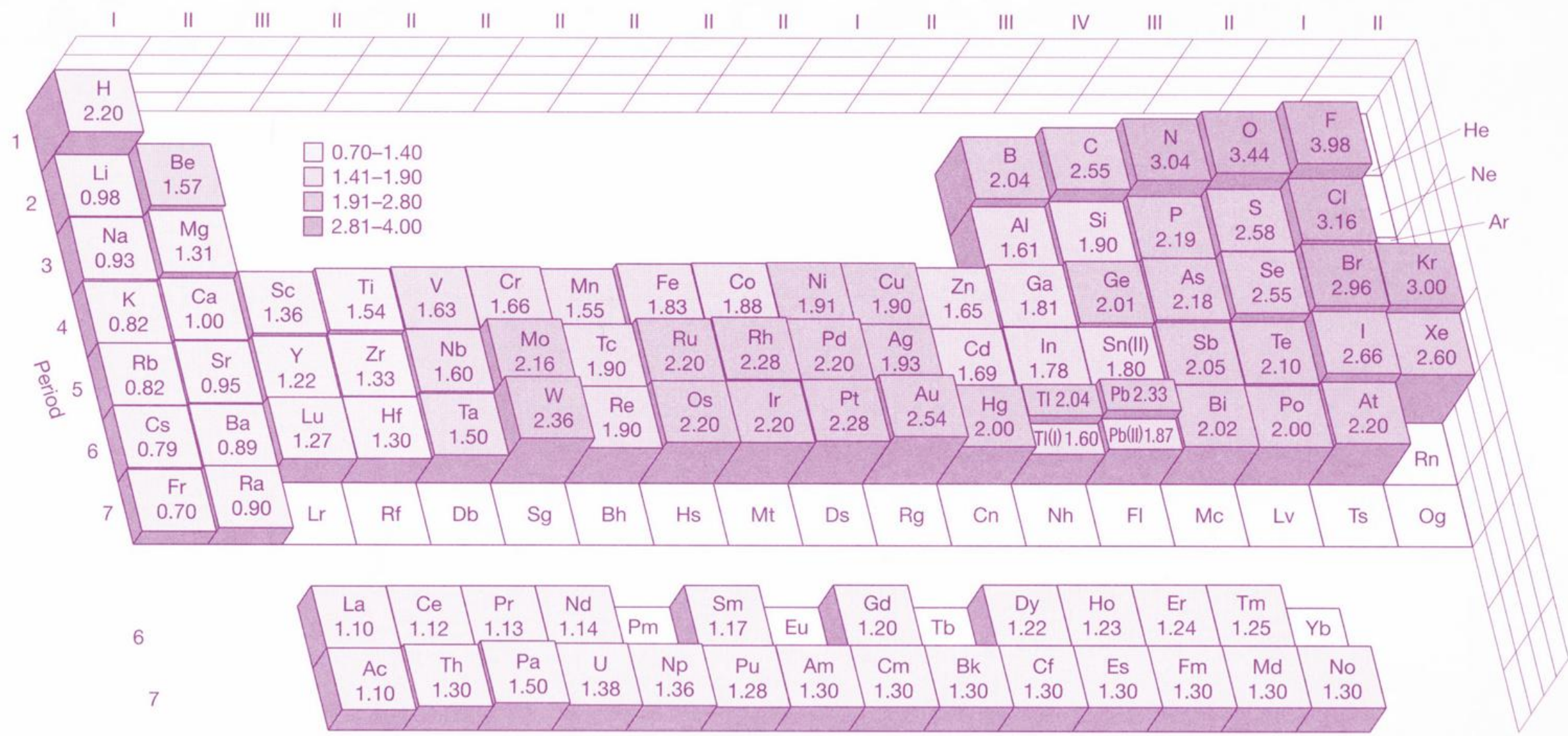


Electronegativity – Take 1

- the tendency of an atom in a covalent bond to attract electron density
- originally conceived by Linus Pauling
 - used bond dissociation energies (E_d) to quantify the extent to which the energy of a bond of two different atoms (A–B) was larger than the geometric mean of the A–A and B–B bond energies

$$|\chi_A - \chi_B| = \sqrt{E_d(AB) - \frac{E_d(AA) + E_d(BB)}{2}}$$

Electronegativity – Pauling Values



Electronegativity – Take 2

- the tendency of an atom in a covalent bond to attract electron density
- Pauling's electronegativity values (χ_P) are the most common
- others conceive of electronegativity in terms of the energy changes that result when charge is added or taken away from the species

Mulliken electronegativity: $\chi_M = \left(\frac{dE}{dq} \right)_{q=0}$

$$\chi_M = \frac{\Delta E}{\Delta q} = \frac{IE - (-EA)}{2}$$

Electronegativity – Take 3

Allred and Rochow extended this idea of electronegativity as *force* to utilize Z_{eff} .

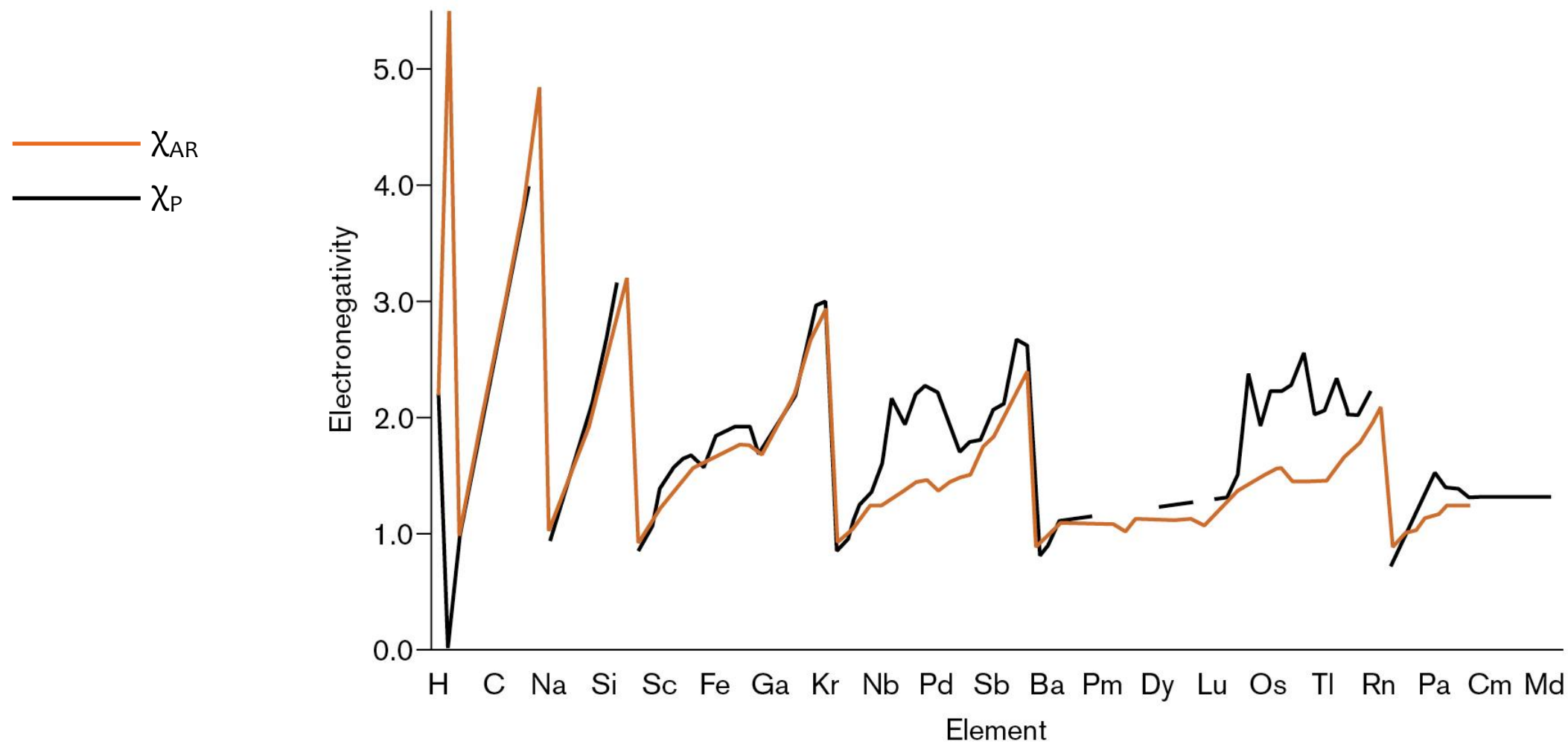
- the higher the electrostatic force at the atom's surface, the greater the tendency of that atom to attract electrons when in a bond

$$F \propto \frac{(Z_{\text{eff}})}{r^2}$$

$$\chi^{AR} = \frac{(0.359)(Z_{\text{eff}})}{r_{\text{cov}}^2} + 0.74$$

covalent radius (in Å) is used since χ is a property of an atom in a bond

Comparison of the Values



Electron Affinities

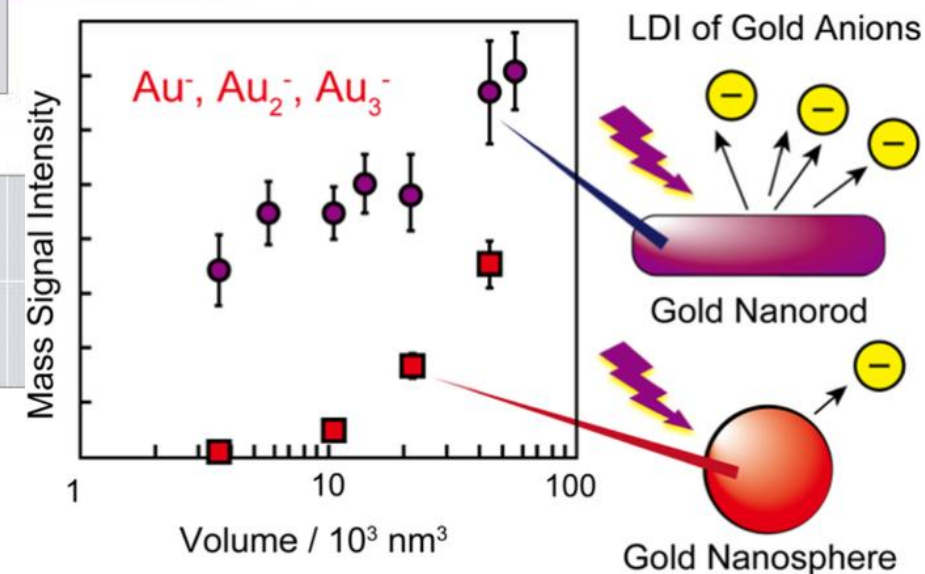
1																	18	
1	H -72.8																He ≥0	
2	Li -59.6	Be ≥0											B -27.0	C -121.8	N ≥0	O -141.0	F -328.2	Ne ≥0
3	Na -52.9	Mg ≥0											Al -41.8	Si -134.1	P -72.0	S -200.4	Cl -348.6	Ar ≥0
4	K -48.4	Ca -2.4	Sc -18	Ti -8	V -51	Cr -65.2	Mn ≥0	Fe -15	Co -64.0	Ni -111.7	Cu -119.2	Zn ≥0	Ga -40	Ge -118.9	As -78	Se -195.0	Br -324.5	Kr ≥0
5	Rb -46.9	Sr -5.0	Y -30	Zr -41	Nb -86	Mo -72.1	Tc -60	Ru -101.0	Rh -110.3	Pd -54.2	Ag -125.9	Cd ≥0	In -39	Sn -107.3	Sb -101.1	Te -190.2	I -295.2	Xe ≥0
6	Cs -45.5	Ba -14.0	La -45	Hf ≥0	Ta -31	W -79	Re -20	Os -104.0	Ir -150.9	Pt -205.0	Au -222.7	Hg ≥0	Tl -37	Pb -35	Bi -90.9	Po -180	At -270	Rn ≥0
7	Fr	Ra	Ac	Rf	Db	Sg	Bh	Hs	Mt	Ds	Rg	Uub	Uut	Uuq	Uup			
Lanthanides			Ce	Pr	Nd	Pm	Sm	Eu	Gd	Tb	Dy	Ho	Er	Tm				
Actinides			Th	Pa	U	Np	Pu	Am	Cm	Bk	Cf	Es	Fm	Md				

≥0 kJ/mol

-348.6 kJ/mol

Mass Signal Intensity

Au⁺, Au₂⁺



- 1.) Extends on the lanthanide contraction argument to provide a deeper explanation.
- 2.) As the effective nuclear charge increases, the velocity increases, which causes the radii of s-orbitals to contract.

Inert-Pair Effect

Tl^{3+} Combined Ionization Energies: 5438 KJ/mol

In^{3+} Combined Ionization Energies: 5082 KJ/mol

Gold is more electronegative.

Learning Outcomes

- 1.) Explain the trends in ionization energy and atomic radii using effective nuclear charge and energy level, n .
- 2.) Understand the different approaches for calculating electronegativities.
- 3.) Identify where relativistic effects become pertinent on the periodic table.
- 4.) Explain the concept of the inert-pair effect.
- 5.) Calculate effective nuclear charge using Slater's rules.

i. Calculate the Z_{eff} for a 3p electron in Sulfur.

ii. What is the most common oxidation state for Bi?

iii. Why is reduction MORE favored for Pb^{4+} than Sn^{4+} ?